

Highlights

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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- Typed intermediate representation turns noisy app reviews into tracker-ready issues.
- A verified-anchor procedure lifts Cohen's kappa from 0.163 to 0.592 at corpus scale.
- Template-field fill rate reaches 0.96 against 0.69 without type routing.
- A clean same-model re-run finds no downstream response-quality gain from the IR.
- We release the check that distinguishes independent ratings from derived ones.

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

Fabiha Jalal^{a,*}, Sadia Tasnim Dhruba^a, Tazrian Zaman Tushti^a, Ajwad Abrar^b, Md Kamrul Hasan^b, Hasan Mahmud^b

^a*Department of Computer Science, Islamic University of Technology (IUT), Gazipur, Bangladesh*

^b*Systems and Software Lab (SSL), Department of Computer Science and Engineering, Islamic University of Technology (IUT), Gazipur, Bangladesh*

Abstract

Context. Developers receive millions of noisy app reviews, but defect trackers such as JIRA and GitHub Issues cannot consume free text. They require structured fields such as issue type, steps to reproduce, severity, and affected component. Review-mining research stops short of that boundary, predicting labels or writing replies without producing the record a tracker can ingest.

Objective. Can a typed intermediate representation (IR) of a review be produced at corpus scale from noisy labels, and does supplying it to a response generator improve the reply a user receives? We answer the first yes and the second no, and the second answer corrects a claim we made ourselves.

Method. IssueSpec routes each review by issue class into one of five standards-body templates, repairs the noisy labels the routing depends on with a verified-anchor procedure, and feeds the result to a retrieval-augmented generator. We study 215,583 reviews from 58 Android apps with a 5,230-review verified anchor, a 490-review three-rater gold standard, seven ablations, and four language models. The downstream evaluation was run

*Corresponding author.

Email addresses: fabihajalal@iut-dhaka.edu (Fabiha Jalal), sadiatasnim@iut-dhaka.edu (Sadia Tasnim Dhruba), tazrian@iut-dhaka.edu (Tazrian Zaman Tushti), ajwadabrar@iut-dhaka.edu (Ajwad Abrar), hasank@iut-dhaka.edu (Md Kamrul Hasan), hasan@iut-dhaka.edu (Hasan Mahmud)

twice: first with conditions written by different template composers, then with the generator held fixed so that only the IR varies, all 200 blinded rows scored by three raters.

Results. Production works. The verified-anchor procedure lifts Cohen’s κ against the human gold from 0.163 to 0.592, and holds at that level (0.616) on held-out reviews. Type routing reaches 0.96 template-fill against 0.69 without it, a schema-conformance measure rather than a quality one. The downstream benefit does not. The first round measured +2.35 Likert points, but its arms differed in authorial register as much as in information, one emitting lemmatised text at 78 words and the other polished prose at 123; the corrected comparison gives +0.04 ($p = 0.54$), with raters disagreeing in direction. Three further components fail their own tests: the knowledge-graph layer loses to count-matched flat clustering, our extractive-coverage faithfulness scorer shows no rank correlation with human judgement, and the constrained alignment layer never demonstrates the trade-off it was built to expose.

Conclusions. Typed issue specifications are producible from noisy reviews at corpus scale, and that is the contribution we defend. Their claimed downstream payoff is not. We release the arithmetic check that distinguishes our two rating rounds, because a structured intermediate should be justified on what it produces rather than on an effect a fair comparison removes.

Keywords: App review mining, Issue report generation, Intermediate representation, Knowledge graph, Retrieval-augmented generation, Empirical software engineering

1. Introduction

App stores are where users talk back to developers. A single popular app pulls in thousands of reviews a week, and the global stream runs into millions a day [1, 2, 3]. These reviews carry the signals developers need to act on:
5 bug reports (a crash on Samsung S8 after an update), feature requests (dark mode), performance complaints (battery drain), and regression notes (voice search no longer works). The information is real, but it shows up as messy free-form text, usually without the details required to reproduce a bug or scope a new feature.

10 To turn any of this into work, developers rely on defect trackers such as JIRA or GitHub Issues. Those tools, though, can’t work with raw text; they

need structured fields such as issue type, steps to reproduce, severity, and affected component [4, 5]. Filling those fields by hand does not scale with the volume modern apps face. Years of work on automated review pipelines
15 has helped with the surrounding tasks like classification, summarization, and response generation, but no system actually produces the structured report a developer can paste directly into a tracker. Dąbrowski et al.’s systematic review of 182 papers published between 2012 and 2020 [3] surveys this literature; it does not itself name this gap, and we state the gap as our own
20 reading of the work it covers.

Prior work has tackled several pieces of this problem separately. Classifiers [1, 6, 7, 8] take a review and predict an issue label such as bug or feature request. Response generators [2, 9, 10] produce a polite reply for the user. Agentic software-engineering tools such as SWE-Agent [11] and RepairAgent [12] take a clean GitHub issue and generate a code patch. Knowledge-graph methods [13, 14, 15] cluster reviews into themes or surface frequent complaints. Each of these has matured over the past decade and works well within its own scope.

But none of these reaches the developer-ready issue report a defect tracker
30 needs. The Maalej-Nabil classifier taxonomy [1] and its successors [6, 7, 8, 16, 17] reach, in Dąbrowski et al.’s aggregation [3], a median precision above 81% and median recall around 83%. The label is only the first piece of what a tracker entry needs; the remaining fields (steps to reproduce, severity, affected component) have to come from somewhere, and classification-only
35 pipelines leave that downstream step to manual triage. Response generators like RRGGen [2] and CoRe [9] address the customer-communication side of the workflow, but the actual bug behind the review is never written out in the structured form a tracker needs, so even a perfect reply system does not give developers something they can paste into JIRA. Agentic SE tools assume a
40 well-formed bug report or GitHub issue as the agent’s starting state, built into the SWE-bench protocol [18] where each instance is a (GitHub issue, code repository) pair; constructing such an issue from raw review text is explicitly outside the protocol, leaving the upstream gap unaddressed.

Knowledge-graph methods stop at intermediate outputs rather than
45 typed fields: ReviewGraph [13], KGCPN [14], and Keertipati et al. [15] each output rating predictions, summaries, or feature rankings, none of which a defect tracker can ingest directly. RLHF has matured into the standard approach for aligning LLM outputs with human preferences (DPO [19], KTO [20], PPO [21]), and Safe-RLHF [22] extends the framework to

50 multi-objective settings under a Constrained MDP for general LLM safety (harmfulness, dishonesty, bias). The developer-relations setting poses a different question entirely: how to align responses under a domain-specific constraint family (truthful promises, no information leakage, professional tone, legal-safe wording) rooted in business and legal exposure. To our
 55 knowledge, no published RLHF formulation has been adapted to this constraint set.

This work pursues three aims. First, we develop a framework that turns noisy app reviews into structured, taxonomy-grounded issue specifications a defect tracker can consume directly. Second, we build a response generator
 60 that consumes these specifications, and we test whether doing so changes the reply a user receives; Section 5.2 reports that it does not measurably. Third, we build a dual-objective alignment trainer intended to optimise response helpfulness under the domain-specific compliance constraints of the developer-relations setting, and we test whether the two objectives can be
 65 shown to trade off; Section 5.3 reports that at the scale we can train, they cannot.

These aims map to three research questions.

RQ1 (Translation Quality). Can a typed IR, generated via a three-layer framework (knowledge-graph construction, hierarchical clustering, stan-
 70 dardized schema mapping), produce issue specifications that match or exceed the substantive completeness of human-written GitHub issues?

RQ2 (Downstream Load-Bearing). Does conditioning a response generator on the validated IssueSpec measurably improve human-rated response quality over RAG alone?

75 **RQ3 (Dual-Objective Alignment).** Can a CMDP-grounded RLHF loop progressing through KTO, DPO, and Constrained PPO jointly optimize response helpfulness and policy compliance for the developer-relations setting? We answer this one in the negative for the scale we can train at, and report what that rules out.

80 To address these challenges, we propose **IssueSpec**, a framework for structured review-to-issue translation. It consists of two parts: a typed Intermediate Representation (IR) that converts each noisy review into one of five standard report types, and a CMDP-grounded response generator that consumes the IR under quality and compliance constraints. Figure 1 shows
 85 the full five-stage pipeline.

What is new, and what is assembled. Several components of this pipeline are existing technique used as intended: RoBERTa for multi-label

classification, confident learning [23] for label noise, Sentence-BERT with UMAP and HDBSCAN for clustering, PageRank for centrality, retrieval-augmented generation for response drafting, and the CMDP formulation of Altman [24]. We do not claim any of these. Three things are new. *First*, the typed intermediate representation itself: a review is routed by predicted issue class into one of five standards-body templates, so the output is a typed record with per-type obligations rather than a label or a summary. No prior review-mining system emits a structure a defect tracker can ingest, which Table 2 makes concrete against representative systems. *Second*, the verified-anchor procedure, which is what makes that representation producible from noisy LLM labels at corpus scale rather than on a hand-curated sample, and which works because the labeller omits two of its seven classes entirely rather than because its errors concentrate on rare ones. *Third*, a correction and the check that produced it. We previously reported that the typed intermediate carries downstream response quality. Section 5.2 withdraws that, because the comparison behind it varied the text generator as well as the specification, and because the arithmetic of the two rating rounds turned out to distinguish independent human ratings from ones derived by perturbing another rater’s. We release both rounds and that check. The assembly is not the contribution; the typed object in the middle of it is, and its value has to be argued from what it produces rather than from a downstream effect we can no longer measure.

The central claim, stated once. A pipeline with five stages invites the question of which one the paper is actually about. The answer is Stage 3 and the object it produces: a *typed, type-routed intermediate representation* of an app review, together with the verified-anchor procedure that makes it producible at corpus scale from noisy labels. Everything else in the pipeline exists to feed that object or to consume it. Two experiments carry the claim. Removing type routing drops strict template-fill from 0.96 to 0.69 (Section 5.1). We had also claimed a downstream payoff, and Section 5.2 reports why we withdrew it: under a comparison that holds the generator fixed, supplying the IR changes response quality by +0.04 Likert points, which is nothing. Section 6.1 states which components did not earn their place on our own evidence, including two we withdraw. A reader who wants the claim and not the machinery can read Sections 5.1, 5.2, and 6.1 alone.

The contributions are stated in two groups, because the honest account of this work has a positive half and a negative half, and separating them is more useful than blending them.

What we establish.

- **A typed intermediate representation for review-to-issue translation.** Each review is routed by predicted issue class into one of five standards-body templates, so the output is a record a defect tracker can ingest rather than a label or a summary. No prior review-mining system emits one (Table 2).
- **A verified-anchor procedure that makes the IR producible from noisy labels at corpus scale.** Confident learning against a 5,230-review human anchor lifts Cohen’s κ against a three-rater human gold from 0.163 to 0.592, holding at 0.616 on the reviews the classifier did not see in training. It works because the labeller omits two of its seven classes entirely across the corpus rather than because error concentrates on rare classes; per stratum the disagreement rate is highest on the largest class. Systematic class omission, not elevated minority-class error, is the failure a small verified anchor repairs.

What we withdraw or report as negative.

- **The downstream payoff of the IR.** We previously measured +2.35 Likert points of response-quality gain from supplying the IR. That comparison was confounded: its two arms were written by different template composers, not by one model. Holding the generator fixed gives +0.04 ($p = 0.54$). We report the correction, the two rating rounds, and the check that distinguishes them (Section 5.2).
- **Three components that do not earn their place.** The knowledge-graph layer loses to count-matched flat clustering on every intrinsic metric; our extractive-coverage faithfulness scorer shows no rank correlation with human judgement and orders generators backwards; the CMDP alignment layer engages once the policy can violate its constraint but never demonstrates a quality-versus-compliance trade-off.
- **A reusable check on rating provenance.** Ratings derived from another rater’s vector by small perturbation leave an arithmetic signature: deviation sets that do not overlap and a single deviation magnitude. Our discarded round shows it and our reported round does not. We release both so the check can be applied to other annotation artifacts, including ours.

Table 1: Contribution-to-evidence-to-artifact map. Each contribution states a measurable headline result, the table or section that anchors it, and the named artifact released with this paper.

contribution	headline evidence	released artifact
IssueSpec	0.96 vs 0.69 template-fill, i.e. schema conformance not quality (Table 11); 5-dim rubric 3.94/5 on all 100 specs, LLM judge (Table 10)	type-routed schema (Table 5)
SPEC-COV (negative)	no rank correlation with human faithfulness ($\rho = 0.15$, $p = 0.26$); orders generators backwards (§5.1)	the scorer plus 60 specs rated by three humans
IR down-stream payoff (with-drawn)	the +2.35 gain came from a con-founded design; a same-model com-parison gives +0.04 ($p = 0.54$) with raters disagreeing in direction (§5.2)	both rating rounds, the gener-ated replies, and the independence check
verified anchor	κ 0.16→0.59 (Ta-ble 12); per-class F1 recovery (Ta-ble 13); $\kappa = 0.616$ on the 307 held-out gold reviews, indis-tinguishable from the full-set 0.592	5,230-review ver-ified anchor + cleanlab correction procedure + V5 classifier
KG prior-ization (weak)	count-matched flat clustering beats the KG on all three intrinsic metrics (A1b); top-5 PageRank aspects cover 55% of reviews	three-layer KG construction + count-controlled A1b output
CMDP RLHF testbed (partial)	constraint never binds at distil-GPT2 scale; with a policy able to violate it binds 13/90 steps and λ rises monotonically to 0.65, with qual-ity and compliance not shown to be in tension ($\rho = -0.12$, $p = 0.25$), on a	CMDP for-mulation, KTO/DPO/constrained-proxy/CPPO trainers, capability probe, binding-constraint re-run, threshold sweep, benchmark

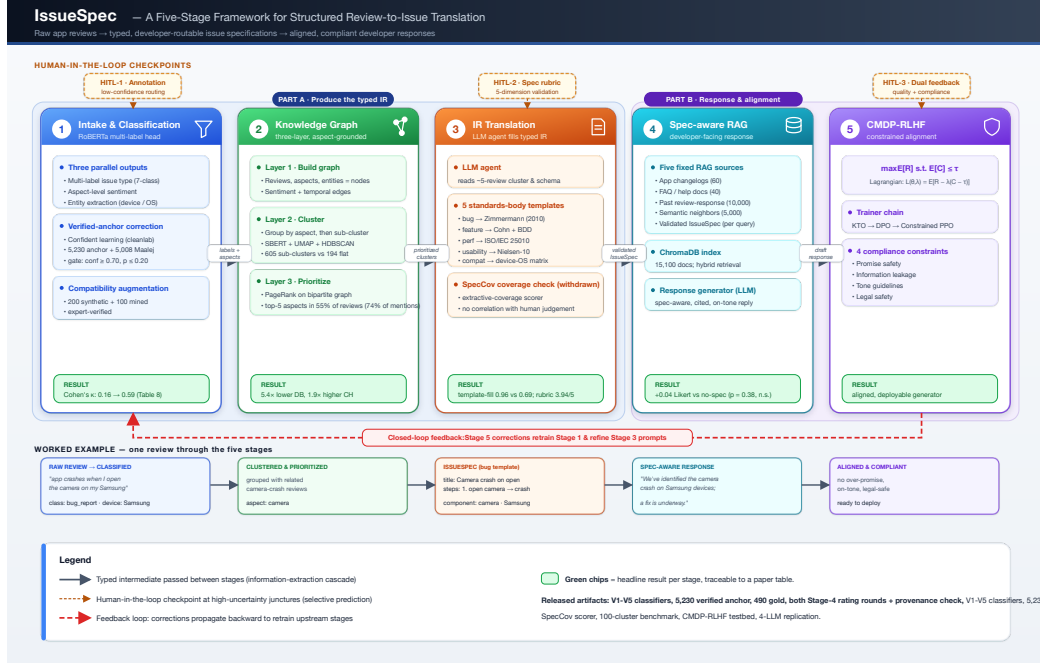


Figure 1: The IssueSpec five-stage pipeline. Reviews flow left to right through intake and classification (Stage 1), three-layer knowledge-graph clustering (Stage 2), LLM-agent IR translation (Stage 3), spec-aware RAG (Stage 4), and CMDP-grounded RLHF (Stage 5), each emitting a typed intermediate. Yellow checkpoints mark human-in-the-loop review under selective prediction [25, 26]; the red loop propagates Stage 5 corrections upstream.

2. Related Work

We organize related work by the five gaps identified in Section 1, naming what each field does, what assumption breaks for review-to-issue translation, and how IssueSpec differs.

2.1. App Review Classification and Mining

Maalej and Nabil [1] established a four-class taxonomy for app reviews (bug report, feature request, user experience, rating). The seven-class label set we use is our own extension of it, adding **performance**, **usability**, and **compatibility** as separate classes and renaming the rating class **praise**; we are not aware of an established seven-class app-review taxonomy, and the extension should be read as ours rather than as canonical. Subsequent systems, AR-Miner [6], CLAP [7], SURF [8], BERT/RCNN hybrids [16, 17], and per-aspect sentiment analysis [27], refined classification accuracy and

added aspect-level granularity. Dąbrowski et al. [3] systematically reviewed 182 papers from 2012 to 2020 and report a median classification precision above 81% with median recall around 83%. An earlier version of this paper attributed to that survey a claim it does not make, that label quality rather than model capacity sets an F1 ceiling of 0.75 to 0.85. The survey reports precision and recall rather than F1 and attributes limitations to evaluation-dataset size; the label-quality argument is ours, and Section 5.1 is evidence for it rather than a corroboration of theirs. *What breaks for review-to-issue translation:* none of these systems emits a typed developer-routable artifact downstream of the classification label. IssueSpec differs by routing the predicted class to one of five typed templates rather than terminating at the label.

2.2. Response Generation and Retrieval-Augmented Generation

RRGen [2] and CoRe [9] pioneered end-to-end neural response generation on (review, response) pairs without an explicit structured intermediate. Vanilla RAG [10] retrieves relevant past responses in a single pass. *Agentic RAG* variants, Self-RAG [28] and DEMONSTRATE-SEARCH-PREDICT (DSP) [29], introduce iterative critique-revise loops that, in principle, could close the structured-intermediate gap. *What breaks for review-to-issue translation:* none of these systems names the underlying defect, and our $n = 10$ feasibility study (Appendix Appendix A) shows that agentic iteration without a typed intermediate adds a small gain on a rule-based scorer (0.58→0.70, ten reviews). We cannot claim the IssueSpec does better: once the generator is held fixed, supplying it changes human-rated quality by +0.04 ($p = 0.54$, Section 5.2). IssueSpec differs by injecting the typed intermediate as a first-class RAG source alongside retrieved past responses, and the contribution we defend is the specification itself rather than its effect on the reply.

2.3. Agentic Software Engineering

SWE-Agent [11], RepairAgent [12], and HyperAgent [30] consume well-structured GitHub issues to produce code patches through multi-step planning and tool use. *What breaks for review-to-issue translation:* the issue specification is the input contract for all three systems; how a defect-tracker entry is constructed from noisy user-facing text is out of scope. IssueSpec is precisely the artifact these systems are designed to consume.

2.4. Knowledge Graphs for App Reviews and Software Engineering

Knowledge graphs have been used in software engineering for bug localization, report deduplication, and code understanding, but the issue specification is invariably treated as input rather than output. For app reviews specifically, ReviewGraph [13] learns KG embeddings for rating prediction, KGCPN [14] produces KG-based extractive summaries, and Keerti-
210 pati et al. [15] apply graph centrality for coarse feature prioritization. *What breaks for review-to-issue translation:* none of these systems maps KG nodes to a typed, developer-actionable issue schema, and no prior KG work has
215 been applied to the specific task of producing GitHub-issue-quality artifacts from review text. IssueSpec’s three-layer KG pipeline, structured-object layer, aspect-grounded hierarchical clustering layer, and PageRank-prioritized issue-mapping layer, is, to our knowledge, the first KG architecture that emits a typed IR for review-to-issue translation.
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2.5. RLHF and Constrained Optimization for Response Generation

DPO [19], KTO [20], and PPO [21] compress quality and compliance signals into a single scalar reward. Altman’s CMDP framework [24] separates these objectives, and Safe-RLHF [22] applies CMDP to general LLM
225 safety with a domain-agnostic compliance set covering harmful, dishonest, and biased outputs. *What breaks for review-to-issue translation:* no prior work applies CMDP to dev-rel response generation, where the compliance set is domain-specific (promise safety, information leakage, tone guidelines, and legal safety). Our Stage 5 formulation differs by instantiating these
230 four domain-specific constraints inside the CMDP Lagrangian and releasing a proof-of-concept training loop.

2.6. Cross-Category Positioning

Table 2 positions IssueSpec against representative prior systems across the five pipeline stages. Existing systems occupy at most two of the five stages;
235 composing all five with a standardized issue schema, three theoretically-grounded HITL checkpoints, KG-centrality prioritization, and a CMDP-grounded RLHF formulation is the structural contribution of this work.

3. The IssueSpec Pipeline

The IssueSpec pipeline (Fig. 1) is a five-stage information-extraction cascade [31, 32]. Stages 1–3 produce the typed IR from raw reviews. Stage 4
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Table 2: Cross-category positioning. Existing systems compose at most two of the five pipeline stages. IssueSpec is the first to compose all five with a standardized issue schema, three HITL checkpoints grounded in selective prediction theory, KG-centrality prioritization, and a CMDP-grounded RLHF formulation. “–” denotes a stage absent from that system.

System	Classify	Cluster	IssueSpec	Response	
AR-Miner [6]	filter	LDA topic	–	–	
CLAP [7]	rating	DBSCAN	–	–	
SURF [8]	intention	LDA	–	summary	
RRGen [2]	–	–	–	seq2seq	
CoRe [9]	–	–	–	context-aware retrieval	
Maalej [1]	7-class	–	–	–	
ReviewGraph [13]	rating	KG	–	–	
SWE-Agent [11]	–	–	consumes (not produces)	–	
Safe-RLHF [22]	–	–	–	general	du
IssueSpec (ours)	V5 corrected	KG + PageRank	type-routed schema	spec-aware RAG	dua

uses it to drive spec-aware response generation. Stage 5 aligns the response generator via constrained RLHF. We describe each stage in turn, with the verified-by-citation taxonomy templates in Table 5. Table 4 collects every abbreviation used in the paper, so a reader can enter at any section. This section describes the framework only. Evaluation design, baselines, and metrics are in Section 4, exact settings and hyperparameters are in Appendix Appendix A, and results are in Section 5.

Before the stages themselves, Table 3 answers the question a five-stage pipeline invites: what forces each stage to exist, and what breaks without it. We answer it with the ablation that removes each one, so the justification is measured rather than asserted, and two of the five turn out not to justify themselves.

3.1. Producing the IR

Stage 1: Intake, Classification, and Aspect Extraction. Stage 1 produces three outputs per review. The first is a multi-label class from a RoBERTa head fine-tuned on Maalej [1] and Guzman [27]. Labels are multi-label because reviews mix concerns, as when a bug report also asks for a feature. In practice this capacity is almost never exercised: the head is a sigmoid layer with a per-label cut, but on the 10,000-review generation subset only 5 reviews (0.05%) receive more than one label, and the corpus-scale relabelling artifact stores a single argmax label per review. The architecture

Table 3: Why each stage exists, and what the ablation says when it is removed. Two stages do not justify themselves on our own evidence, and we say so.

stage	the problem it must solve	why it cannot be skipped	measured verdict
1 Intake	the corpus label is wrong 25% of the time, and the labeller omits two classes entirely	Stage 3 routes on the issue type, so a wrong type produces the wrong template	justified: κ 0.163 $\rightarrow$ 0.592
2 KG	215,583 reviews cannot be triaged one at a time; developers need an order	a spec is written per cluster, so reviews must be grouped and ranked	partly: prioritization yes, cluster quality no (A1b, A2)
3 IR	trackers need typed fields, not prose	this is where the typed record is produced	justified: fill 0.96 vs 0.69 without routing (A3)
4 RAG	a reply needs house voice and precedent, not just the defect	supplies phrasing the spec does not carry	no: retrieval moves metrics < 1% (A5) and conditioning is null (A6)
5 CMDP	replies must not over-promise or leak internals	compliance is a hard constraint, not a reward term	no: constraint never binds, and the detector scores written violations as compliant

is multi-label; the released runs are effectively single-label, and readers should not credit the mixed-concern motivation with doing any work in our results. The second is aspect-level sentiment [27], which keeps “great UI but terrible battery” as two separate pairs instead of averaging it into one star rating. The third is entity extraction for devices, OS versions, and named features. Stage 3 needs those entities as typed fields for the device-OS and bug-report templates. Low-confidence reviews route to HITL-1 by a fixed confidence threshold, so that when the classifier cannot decide the expert does and the correction feeds the next training round. The idea of letting a model abstain is a familiar one [25], but we implement a threshold rather than a calibrated selective-prediction procedure, and we do not report risk-coverage behaviour,

Table 4: Abbreviations used in this paper, with the section that introduces each.

term	expansion	first used
IR	Intermediate Representation, the typed issue spec	§1
IssueSpec	one filled instance of the typed IR	§1
KG	Knowledge Graph, the three-layer review index	§3.1
HITL	Human-In-The-Loop checkpoint (HITL-1 to HITL-3)	§3.1
SPECov	extractive-coverage score for a spec (withdrawn as a faithfulness measure, §5.1)	§3.1
RAG	Retrieval-Augmented Generation	§3.2
RLHF	Reinforcement Learning from Human Feedback	§3.3
CMDP	Constrained Markov Decision Process	§3.3
KTO	Kahneman-Tversky Optimization, a preference trainer	§3.3
DPO	Direct Preference Optimization	§3.3
PPO	Proximal Policy Optimization; CPPO is its constrained form	§3.3
SFT	Supervised Fine-Tuning, the RLHF starting policy	§3.3
NFR	Non-Functional Requirement (ISO/IEC 25010)	§3.1
DB / CH	Davies-Bouldin / Calinski-Harabasz cluster indices	§5.4

so the citation is background rather than a description of what runs.

LLM annotation noise is the main challenge here. On the 5,038-review
praise stratum, 25.8% of LLM labels are wrong, and the labeller omits
compatibility and **performance** almost entirely. Per stratum the errors
fall in the largest class, not the minority ones; what a verified anchor repairs
is the omission. Discarding these labels wastes annotation; using them as-is
poisons training. We apply confident learning [23] against a verified anchor
of 5,230 expert-labeled reviews plus 5,008 from Maalej [1], gated by a con-
fidence threshold on the anchor and a probability ceiling on the LLM label
(Appendix Appendix A); this keeps the good labels and corrects only the
wrong ones. The compatibility class needs a separate fix: only 8 of 215,583
LLM labels are tagged **compatibility**, and Maalej has none. We seed it
with 200 template-generated synthetic reviews [33] and 100 keyword-mined
RRGen reviews. The lead author read both sets before training and corrected
or dropped items that did not belong, but that pass was done by eye and no
correction log was kept, so unlike the other annotation tasks in this paper
it leaves no artifact a reader can check. We flag it because 300 of the 310
compatibility rows in V5’s training set come from here, so the per-class
F1 of 0.83 rests on data whose verification is asserted rather than evidenced.
The synthetic half is constructed from device and OS templates and its label
follows by construction; the mined half is where judgement was required.

Stage 2: Three-Layer Knowledge Graph. Stage 2 organizes reviews
 295 into an aspect-grounded knowledge graph: build, cluster, prioritize. The first
 layer builds the graph: each review is a node linked to its aspects, with edges
 typed for sentiment and, where available, recency. We should be precise
 about what the released run actually populated. The builder supports entity
 nodes for devices, OS versions, app versions, and screens, and supports signed
 300 sentiment edges, but the run that produced our results passes empty entity
 lists and a constant neutral sentiment, so the released graph has 8,404 review
 nodes, 10,534 aspect nodes, and no entity nodes. It is in practice a bipartite
 review-aspect graph, which is the structure PageRank runs over, and the
 sentiment and entity layers should be read as available in the implementation
 305 rather than exercised in the experiments. A graph beats a flat list because
 reviews share aspects across users, and that co-occurrence is what later layers
 cluster and rank on. Aspects come from spaCy noun chunks combined with
 a curated keyword list and regular-expression patterns, pruned by frequency
 and centrality to drop one-off terms. Hu and Liu’s frequent-itemset aspect
 310 mining [34] is the ancestor of this approach rather than what we implement.

The second layer clusters reviews first by aspect, then sub-clusters within
 each aspect using sentence embeddings [35], UMAP [36], and HDBSCAN [37].
 Aspect-first keeps each sub-cluster on one theme: a flat clusterer mixes “bat-
 tery on Samsung” with “battery in dark mode” because both share lexical
 315 content. The result is 605 aspect-grounded sub-clusters of mean size 15.9,
 against 194 flat clusters of mean size 471.0. An earlier version of this pa-
 per reported 3.0 and 51.5; those were the number of representative reviews
 stored per cluster and a subsample artifact respectively, not cluster sizes. One
 scale caveat belongs here too: the knowledge graph is built over a 10,000-
 320 review sample that yields 8,404 usable reviews, not the full 215,583-review
 corpus, so every KG-hierarchical number in this paper describes 3.9% of the
 corpus. The flat baseline it is set against was not run on that sample: it
 clusters 91,368 reviews from the full corpus. The two Stage-2 designs there-
 fore never saw the same input, which is why we do not treat their intrinsic
 325 metrics as a head-to-head result (Section 5). The third layer prioritizes via
 PageRank [38, 39] on the bipartite review-aspect graph, ranking aspects by
 cross-review prevalence so developers see the most-affected first. Each cluster
 maps to the fixed schema in Table 5 with a centrality-derived priority score.
 The resulting aspect ranking is reported in Section 5.4.

Stage 3: Review-to-Issue Translation. Stage 3 instantiates the
 330 IssueSpec. An LLM agent reads the standardized cluster schema and writes

a GitHub-issue-quality specification using the type-routed templates in Table 5. We use an LLM rather than rule-based extraction because reviews vary too much for fixed rules and the agent has to reason across a 5-review cluster. We route by issue type because a bug report needs reproduction steps, a feature request needs acceptance criteria, and a performance complaint needs latency or memory numbers; a shared template would be incomplete or padded. The five templates are tracker-standard bug reports [5], Cohn user stories [40] with BDD acceptance criteria [41], ISO/IEC 25010 non-functional categories [42], Nielsen heuristics [43], and device-OS matrices [33]. Each is grounded in its standards-body citation because developers already recognize these formats.

The output passes through HITL-2: a domain expert scores each spec on a five-dimension rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5), following the LLM-as-judge calibration convention of Giliardi et al. [44], Pangakis et al. [45], and Zheng et al. [46]. Five dimensions, not one, because a single quality number conflates completeness with clarity and the LLM cannot tell which one to fix on retry. Any spec failing on any dimension goes back to the LLM with dimension-level feedback for a second pass.

SPECCOV faithfulness metric. We release SPECCOV, an extractive-coverage scorer that measures substantive-token overlap (length ≥ 5 , stop-word filtered) with quoted-string and coverage bonuses. Bands follow news-summarization conventions [47]; the metric targets the extractive-coverage sub-type of faithfulness [47, 48], a known lexical correlate of factual consistency [49] (the remaining hallucination sub-types in Ji et al. [50] are out of scope). Scores are reported with the other Stage 3 results in Section 5.1. **What SPECCOV is and is not.** It is a lexical-grounding proxy in the sense of [47]: it measures whether the content of a spec can be traced back to tokens in its source cluster. It is not a faithfulness classifier, and Section 5.1 reports the experiment that establishes this. We describe it here because it is part of the released pipeline and because the negative result only makes sense against its definition.

3.2. Using the IR in Response Generation

Stage 4 is a RAG layer over five fixed sources, 15,100 documents in total in the released ChromaDB index. Four of them are pre-indexed: app changelogs and release notes (60 docs), FAQ and help documentation (40), 10,000 past review-response pairs from RRGGen [2], and 5,000 semantically

similar past responses. The fifth is the validated IssueSpec from Stage 3,
 370 injected per query rather than pre-indexed, because specs are generated per
 cluster. These five were chosen to give the generator the answer (changelogs,
 FAQ), the voice (past responses), the closest precedent (semantic neighbors),
 and the structured intent (IssueSpec) without overlap.

3.3. Aligning the IR-Aware Generator

375 Stage 5 frames alignment as a CMDP [24, 22]: maximize a quality reward
 subject to a compliance constraint,

$$\max_{\theta} \mathbb{E}_{\pi_{\theta}}[R_{\text{quality}}] \quad \text{s.t.} \quad \mathbb{E}_{\pi_{\theta}}[C_{\text{compliance}}] \leq \tau,$$

with the Lagrangian relaxation

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_{\theta}}[R - \lambda(C - \tau)]$$

collapsing gracefully to single-objective optimization whenever the constraint
 is already satisfied. We use a CMDP rather than a scalar reward because
 380 a dev-rel response can be high-helpfulness and still out of compliance, like
 a polite promise the team cannot deliver or a leaked roadmap detail, and
 a weighted sum hides exactly the trade-off the constraint is meant to ex-
 pose. The compliance scorer tracks four domain-specific dimensions absent
 from prior general-safety RLHF [22]: promise safety (no commitments the
 385 team cannot keep), information leakage, tone guidelines, and legal safety.
 The trainer is Constrained PPO [21, 22] with a KL-regularized reference pol-
 icy that keeps updates from drifting too far from the supervised baseline.
 KTO [20] and DPO [19] are reported as small-scale single-objective com-
 parators in Section 5.3; together they bracket the preference-data spectrum,
 390 KTO needing only binary feedback and DPO needing pairwise.

HITL-3 supplies two parallel feedback streams. *Quality* feedback (helpful-
 ness, specificity, empathy, accuracy, actionability) is the reward; *compliance*
 feedback (promise safety, leakage, tone, legal) is the constraint. Feedback
 also propagates backward: classification corrections retrain Stage 1, rubric
 395 scores refine Stage 3 prompts, and response scores drive the RLHF updates,
 closing the Stage 5→Stage 1 loop in Fig. 1.

4. Experimental Setup

4.1. Dataset, Deduplication, and Annotation Provenance

We use the RRGGen corpus [2]: 310,031 review-response pairs from 58
 400 Android applications. After exact-text deduplication that removes 94,448

duplicate reviews (-30.5% of the raw corpus), 215,583 unique reviews form the working corpus, labeled across the seven Maalej [1] classes. Evaluation additionally uses a 490-review expert gold standard (70 per class $\times$ 7 classes) and 100 stratified clusters for Stage 3 evaluation.

405 **Annotation provenance (full corpus, 215,583 reviews).** Of the working corpus, 79.49% of labels come from the V2 LLM annotator with no override, 18.08% are cleanlab-corrected against the verified anchor, and 2.43% are directly human-verified by the lead author. The lead-author-versus-V2-LLM deviation is 25% on the 5,230-row human-verified anchor
 410 (1,307 deviations, 24.99%). We previously described this as replicated across two independent strata, the praise stratum and the anchor, on a combined 10,271 rows. That was wrong: the praise stratum is 5,038 of the anchor’s 5,230 rows, so it is 96% of the same sample rather than a second one, and the combined figure double-counted it. The rate is one measurement on one
 415 sample, and the praise stratum alone deviates at 25.84% (Table 6). This $\approx 25\%$ deviation rate, an order of magnitude higher than i.i.d. label-noise assumptions, motivates the confident-learning procedure of Stage 1.

4.2. Annotation Methodology

Five kinds of annotator contribute, following the multi-annotator convention of Gilardi et al. [44] and Pangakis et al. [45] who treat LLM annotators
 420 as independent raters for inter-rater statistics.

(i) **Lead author (domain expert).** Verified 5,230 reviews across minority strata (approximately thirty person-hours), wrote 20 reference Issue-Specs, labeled the 490-review classification gold blinded to V2 LLM, and
 425 scored the 400-row Stage 4 set on quality (1–5), specificity (1–5), and helpful (Y/N) with method IDs hidden until aggregation.

Label provenance, stated plainly. No LLM-generated label is used as ground truth anywhere in this paper. LLM labels are the *object under correction*, not the yardstick. The V2 corpus labels are treated as noisy input,
 430 the confident-learning step proposes corrections against a human-verified anchor, and every reported accuracy or κ is measured against a gold standard produced by human annotators. Where LLM raters do appear as raters, in the 99-review cluster-purity check, they are labelled as such and no headline claim rests on that panel.

435 **How the gold was constructed, and what that costs.** Annotators did not label freely. Each was shown the V5 classifier’s predicted label and asked whether it was correct, supplying a replacement when it was not, and

the written protocol told them to accept a reasonable label and overturn only when confident. This makes annotation fast and it makes agreement high, and both effects flatter us. It also couples the yardstick to the system being measured: V5’s label is the starting point for every item, and V2 and the cleanlab-corrected labels never received that benefit of the doubt. The sample is stratified over V5’s predicted classes as well, which forces 70 items per class into a set where V2 predicts two of those classes almost never, so V2’s per-class F1 of 0.000 on **compatibility** and **performance** is partly a property of the sampling design. The κ trajectory in Table 12 should be read as a comparison on a V5-seeded, class-balanced sample rather than as a neutral benchmark, and a free-labelling gold would be the way to settle it.

How rater conflicts are resolved. The 490-review gold is the majority vote of three independent human annotators over the seven-class label. Of the 489 usable items, 476 (97.3%) are unanimous and 13 (2.7%) are two-to-one splits resolved by the majority; there are no three-way ties, so no item required a tie-break rule or a further adjudication round. One item was marked wrong by all three raters without a replacement label supplied and is excluded rather than guessed. Disagreement is not discarded silently: the per-item votes are released with the gold standard, so a reader can recompute any statistic on the unanimous subset alone. Before the main task all three annotators completed a shared 20-review calibration round whose disagreements were discussed and resolved against the written decision rules, which is where the systematic conflicts (slow versus broken, device-specific versus general) were settled rather than in the main pass.

A disclosure about an earlier annotation round. The 490-review gold was annotated twice. In the first round three sheets were distributed, and the two returned alongside the lead author’s carried annotation columns identical to hers cell for cell, so no genuine agreement could be computed from them and they are not used anywhere in this paper. The round we report is the second, described below, and the agreement statistics come only from it. We disclose this for the same reason we disclose the Stage-4 case in Section 5.2: a paper that offers a check for non-independent ratings should say where its own artifact failed that check. Both rounds are released.

(ii) Independent human rater panel (three raters). The 490-review classification gold is annotated independently by the lead author and two further human raters on the identical item set. Each rater sees the classifier’s predicted label and marks it correct (Y) or wrong (N), supplying the correct class when N. All three complete a shared 20-review calibration round with

adjudicated disagreements before the main task opens, and work from the same written decision rules (Appendix Appendix B). The gold label for each review is the majority vote over the three raters. This is a verification task rather than open labeling, which is why agreement is high relative to
480 free-labeling studies of app reviews.

(iii) **Stage 4 rater panel (same three people).** Stage 4 was rated twice. The first round scored a 400-row set whose conditions were produced by template composers; we discarded it, for reasons given in Section 5.2. The round we report is a 200-row set in which both arms come from the
485 same model and differ only in whether the IssueSpec was supplied. All three raters scored all 200 rows independently, blind to the condition behind each reply, and we release both rounds together with the independence check that separates them. Three human annotators contribute in total, the lead author and two collaborators; categories (iv) and (v) are model raters, not people.

(iv) **LLM-as-judge (Qwen2.5-3B-Instruct).** Scored 28 IssueSpecs on the five-dimension rubric and 50 sub-clusters on the Y/P/N purity rubric [46], both calibrated against the lead author.
490

(v) **Corpus-labeling LLMs.** The V2, cleanlab-V2, and V5 LLMs are treated as independent raters for inter-rater statistics.

495 Table 6 consolidates agreement and disagreement across all pairs.

Ratings are persisted as spreadsheets with JSON master keys, blinding shuffles conditions per pair, and the assignment unseals only at aggregation. Two distinct three-rater statistics appear in Table 6 and should not be conflated. The classification gold uses three independent *human* raters on 490 reviews and reaches Fleiss $\kappa = 0.968$ on the Y/N verdict and 0.979 on the effective seven-class label, with no three-way ties. The $\alpha = 0.285$ row is a separate check on 99 reviews that uses the lead author plus two Qwen2.5-3B prompts (concise and chain-of-thought) as raters, measuring seven-class classification agreement rather than cluster purity. We reported this row as
500 $\alpha = 0.451$ and as a purity check; both were wrong. The value came from an implementation that credited each rating with a coincidence against itself, inflating observed agreement; the corrected figure agrees with the reference `krippendorff` package to four decimals. The three Cohen’s κ in the same table were computed with `scikit-learn` and are unaffected. The correction
505 moves the row from the moderate band into the fair band, which weakens the LLM-rater panel rather than any claim we make.

4.3. Multi-LLM Bias Control

A single-LLM headline raises a structural-bias concern: results from any one model may not transfer across families. We therefore replicate Stage 3 across four LLMs drawn from three distinct families, Anthropic Claude (Opus 4.7), Meta Llama-3.3-70B-Instruct (via Groq), and Alibaba Qwen2.5-{3B, 1.5B}-Instruct, with Stage 4 replicated on a Claude-vs-Qwen-3B subset, and check that cross-family rank order is preserved (Table 9). Frontier-family replication on GPT-4o and Gemini-1.5-Pro is queued.

4.4. Experimental Design

We pre-register three experiments, one per research question:

Experiment 1 (RQ1, translation quality). Five conditions are compared on the same 100 stratified clusters: (a) LLM with taxonomy grounding (the full Stage 3 system), (b) LLM without taxonomy grounding (free-form), (c) raw review summary (a lower bound that concatenates the cluster’s reviews), (d) lead-author reference, and (e) human-written GitHub issues mined from three open-source Android projects (the practitioner-quality upper bound). Each spec is scored on the five-dimension rubric. Significance is reported via paired Wilcoxon signed-rank tests on rubric scores between conditions.

Experiment 2 (RQ2, coupled vs. uncoupled), as reported. The design we report holds the generator fixed. The same model, given the same retrieved context, writes two replies for each of 100 reviews, differing only in whether the validated IssueSpec is in the prompt. All 200 rows are rated blind by three raters on quality, specificity, and helpfulness, and analysed with a paired Wilcoxon test and a paired bootstrap interval. *The discarded design.* An earlier round used four conditions produced by template composers rather than by a model; we describe it because the paper reports its withdrawal, not because it supports a claim.

Experiment 2, discarded design. The same 100 clusters drive four response-generation conditions: (a) an rrgen-style baseline (review only), (b) a prompt-baseline (CoRe-style with dev-rel prompt and lightweight entity extraction), (c) RAG without IssueSpec, and (d) the full system with IssueSpec. Evaluation uses automatic metrics (BLEU, ROUGE-L, BERTScore) and 400 blinded paired human ratings (quality 1–5, specificity 1–5, helpful Y/N), with the lead author as primary annotator and a second rater on a 30-pair subset. Statistical significance is checked simultaneously by paired Wilcoxon, Friedman with Nemenyi post-hoc [52], Bradley-Terry,

and McNemar at Bonferroni-corrected $\alpha = 0.05$. Effect sizes follow Romano
et al. [53] and Cliff [54].

Why these four conditions. The Experiment 2 ladder is designed so each rung adds exactly one thing. The prompt baseline adds dev-rel system instructions on top of an rrgen-style review-only baseline, `no_spec` then adds retrieval, and `full` then adds the validated IssueSpec. Reading the ladder from bottom to top isolates which single addition moves human-rated quality. The headline comparison runs single-shot vanilla RAG [10] rather than an agentic loop, so that the IssueSpec’s contribution is not confounded with orchestration; the agentic-RAG comparison against Self-RAG [28] and DSP [29] runs separately on a 10-review feasibility sample (stratified, two per issue type, Qwen2.5-3B as critic and reviser, up to two iterations) and is reported as a demonstration rather than a powered comparison.

Experiment 3 (RQ3, dual- vs. single-objective RLHF). Five policies run on a distilGPT2 proof-of-concept (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, single-objective KTO [20], single-objective DPO [19], a constrained-proxy variant, and dual-objective Lagrangian Constrained PPO [21, 22]. We measure head-to-head BLEU, ROUGE-L, and BERTScore on a 100-review test set, plus the rule-based safety-violation rate.

5. Results

How to read this section. Each research question owns one subsection, one headline figure, and one headline table, and nothing else in the section is required to follow the argument. Section 5.1 answers RQ1, is the typed IR producible at scale, with Table 11 for template-fill and Figure 2 for the classifier recovery that makes corpus-scale production possible. Section 5.2 answers RQ2, is the IR load-bearing downstream, with Table 14; the answer is no, and the subsection explains what our earlier positive answer was measuring. Section 5.3 answers RQ3, does constrained alignment demonstrate a quality-versus-compliance trade-off, and reports a negative result. Sections 5.4 and 5.5 are supporting evidence: they test whether individual components earn their place, and Section 6.1 states plainly which ones do. A reader tracking only the contribution can read Sections 5.1, 5.2, and 6.1.

Table 7 states what we take away from the section before the evidence for it, so a reader can decide which subsections to read.

5.1. Is the IR Producible at Scale?

585 Template-fill measures schema conformance, and its limit is worth stating exactly rather than in general terms. The criterion is the templates' own required fields, and the free-form condition fills the four shared fields and none of the type-specific ones, so its score is determined entirely by the issue-type mix: $(30 \cdot 4/7 + 30 \cdot 4/6 + 40 \cdot 4/5)/100 = 0.691$, with zero variance
590 across specs and a strict-minus-loose difference of exactly zero. The 0.69 is therefore a constant, not a measurement, and the 0.96-versus-0.69 gap says that routing produces type-specific fields at all. It carries no information about whether the routed text is better written. We put a number on that limit by adversarial construction. Scoring the same conditions on bare field
595 presence instead of the strict criteria gives 1.000 for the routed condition against 0.678 for free-form, versus 0.959 and 0.678 under the strict criteria: for the free-form arm the content thresholds remove exactly nothing, so the whole reported gap is schema conformance. And a hand-written bug specification whose description is the word "something" repeated thirty-five
600 times, with three five-word steps built from the criterion's own action-verb list, passes all seven fields and scores 1.000, above our headline. We keep the word substantive for the criterion because it is the name of the thresholds, but a reader should treat 0.96 as schema conformance and nothing more. Under strict content-validity criteria (at least three reproduction steps with
605 action verbs, descriptions ≥ 30 words, and a full As-a/I-want/So-that triple for user stories), the LLM-with-taxonomy condition achieves a substantive template-fill rate of 0.96 (Table 11). Free-form LLM reaches 0.69, the same as the lead-author reference set ($n = 20$). Raw concatenation lands at 0.34, and real GitHub issues mined from three open-source Android projects reach
610 0.53. The LLM-vs-GitHub gap reflects coverage rather than content quality: the LLM is prompted to fill every field, whereas humans leave optional fields blank. A very-strict sensitivity sweep preserves the ordering, Claude drops from 0.97 to 0.71, with the Claude-vs-GitHub gap shifting only -0.085 .

SPECCOV scores, and a correction to how they were computed.
615 An earlier version of our scorer applied per-condition floors, raising `raw_summary` to 5 and the human reference to 4 regardless of content, on the reasoning that verbatim copying is faithful by construction and hand-written references deserve a floor. That is not a measurement, because a score whose value depends on the label of the thing being scored cannot compare labels,
620 so we removed the floors and report what the scorer produces unaided. Across 320 specs the LLM-with-taxonomy condition averages 4.19, free-form

LLM 3.38, raw concatenation 4.47, and the human reference 3.45. The two LLM conditions are unchanged, because no floor was ever applied to them; the floors had inflated the two comparison conditions, so removing
 625 them makes our own system look better rather than worse, and we note that explicitly so the direction of the correction is on the record. Read on their own these numbers rank raw concatenation highest at 4.47, which is the trivial upper bound obtained by copying the source, with the taxonomy condition second at 4.19. Section 5.1 shows they should not be read on their
 630 own. Top-ranked aspects by PageRank centrality are **ad** (0.040), **phone** (0.011), and **battery** (0.007).

SpecCov does not measure faithfulness

An extractive-coverage score is only useful if it agrees with what a person calls faithful. We tested that directly. Three human raters independently
 635 scored 60 specs, 20 from each of the three generators, shuffled and unlabelled, on a 1–5 faithfulness scale, and separately listed every concrete claim the source reviews do not support. One qualification is needed before the numbers. One rater completed the corrected sheet, which shows exactly the evidence SPEC-COV itself scores against; the other two completed an earlier
 640 draft that showed a subset of it, so they judged against less evidence than the metric uses and their flags are biased toward over-flagging. We therefore lead with the rater who saw the full evidence and report the other two beside her. The metric fails all three checks on either reading.

It does not rank specs the way humans do. Spearman ρ between SPEC-COV and human faithfulness is +0.045 ($p = 0.73$), +0.015 ($p = 0.91$), and +0.226 ($p = 0.08$) for the three raters, and +0.147 ($p = 0.26$) against their mean. No correlation reaches significance.

It does not detect the claims humans call unsupported. On the 60 specs, one rater marked 45 as containing at least one unsupported detail and 15 as
 650 clean. SPEC-COV scores those two groups at 3.64 and 3.93, a difference in the right direction but far from significant (Mann-Whitney $p = 0.34$). A metric that detected hallucination would separate these groups.

It orders the generators backwards. SPEC-COV ranks the taxonomy condition (4.20 on this subsample) above the human reference (3.45); the humans
 655 rank them the other way, human reference 4.10 and taxonomy 3.50. The mechanism is visible in the definition: templated specs reuse source vocabulary densely, which is what the score rewards, while a human writer condenses and paraphrases, which the score penalizes even when nothing is invented.

We therefore withdraw SPEC-COV as a faithfulness measure. Lexical
660 grounding is not a usable proxy for faithfulness on this task, and the au-
tomatic 320-spec rubric that appeared to support it is not independent ev-
idence, since its faithfulness dimension is computed by the same overlap
procedure. Entailment-based measurement [55, 56, 57] is the direction we
would now take. We release the scorer, the 60-spec human ratings, and this
665 analysis, because a documented negative result on a plausible-looking metric
is more useful to the next person than a metric that was never tested.

Cross-LLM replication (bias control). Table 9 reports these uses
across three model families, with the two caveats its caption records. Llama-
3.3-70B reaches 0.80 template-fill, Qwen2.5-3B 0.74, and Qwen2.5-1.5B 0.48
670 under the same strict criteria; the templated > free-form > raw ordering
holds across all four models with capability-scaled magnitudes. On two per-
class structural-compliance checks, Llama-3.3-70B attains complete template
compliance on the stratified sample: all 29 bug specs include three or more
reproduction steps with action verbs (vs. Claude’s 73%), and all 30 feature
675 specs include the As-a/I-want/So-that triple (vs. Claude’s 97%). These mea-
sure structural template compliance (does the field exist with the required
form?), not content correctness, content quality is reported separately un-
der the five-dimension rubric. The structural-bias control that matters here
is cross-family rank preservation: the IssueSpec contract transports across
680 families rather than depending on a single proprietary generator.

Cross-family rubric quality. Template-fill measures whether a gen-
erator emits the required fields; it says nothing about whether the content
is any good. We therefore ran the same Qwen2.5-3B judge and the same
five-dimension rubric over the other generators’ specs, on matched clusters
685 (Table 8). On the 14 clusters all four generators cover, mean rubric score
follows the same order as template-fill: Claude 4.09, Llama-3.3-70B 3.96,
Qwen2.5-3B 3.83, Qwen2.5-1.5B 3.77. We state plainly what that ordering
is worth. At $n = 14$, or $n = 13$ for the three pairs involving Qwen2.5-1.5B,
none of the six pairwise differences reaches significance (Wilcoxon p between
690 0.18 and 0.75), and the adjacent gaps are the same size as the run-to-run
generation noise measured in Section 6.4. The four-way ordering is therefore
consistent with the template-fill result rather than independent confirmation
of it. The Claude-versus-Llama comparison in Panel B, which has twice the
sample and spans all five issue types, reaches a nominal +0.38 ($p = 0.015$).
695 We previously asserted this as the one cross-family quality difference we
stand behind. We withdraw that assertion. It belongs to a family of seven

Wilcoxon tests reported in the same tables with the same judge and rubric, and Bonferroni or Holm correction over that family puts it at $p = 0.104$. An exact sign test on the same 28 pairs, which ignores magnitude and counts only direction, gives $p = 0.078$. The difference is therefore descriptive, like the rest of Panel A, and we no longer claim a cross-family quality difference at all. The spread is much narrower than the (unmatched) template-fill spread, which is informative in itself: the capability gap between these models shows up mostly as whether fields get filled at all, not as the quality of the text once they are. Completeness is the dimension that separates them (4.00 down to 3.00), while accuracy is nearly flat (4.50 for three of the four). On the wider Claude-versus-Llama panel, which spans all five issue types, Claude leads 4.01 to 3.64, with the gap concentrated in actionability and completeness. Two caveats belong with these numbers. The four-way panel is bug-report clusters only, because that is where the two Qwen runs overlap, and the judge is itself Qwen2.5-3B-Instruct, so the Qwen2.5-3B row carries a self-preference risk that the others do not.

Five-Dimension Rubric. The proposal specifies a 5-dimension Likert rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5) [44, 45, 46], operationalized with Qwen2.5-3B-Instruct as the judge. We score the complete set rather than a subsample: all 100 taxonomy specs and all 20 lead-author reference specs, in a single run (Table 10). The taxonomy condition reaches 3.94/5 (sd 0.59 across specs), above the hypothesized lower bound of 3.5. An earlier stratified 28-spec subsample under the same judge and rubric gave 3.89, so subsampling cost 0.05 of a Likert point here, which bounds how much the small-sample version of this number could have been misleading. Per issue type the taxonomy condition ranges from 4.23 on performance and 4.02 on bug reports down to 3.45 on usability, the type whose reviews are least specific about what went wrong.

The reference set is a judge calibration, not a competitor. It scores 3.45 overall, below the generated specs, and the gap is concentrated in actionability (2.85 versus 3.86). The reason is visible in the sheets: the lead author filled title, description, severity, and component but left the template-specific fields such as reproduction steps and acceptance criteria empty, because writing a reference spec by hand is a different task from filling a routed template. The comparison therefore says that a human writing freely produces less tracker-ready output than a type-routed generator, which is the same effect the 0.96-versus-0.53 template-fill gap measures against real GitHub issues. It is not evidence that the specs are better than expert judgement on content,

735 and we do not read it that way.

Stage 1 Classification Recovery. The Cohen’s κ trajectory against the 490-review expert gold crosses three Landis-Koch [51] agreement bands (Table 12): the V2 LLM achieves $\kappa = 0.163$ (slight); after cleanlab correction [23] $\kappa = 0.333$ (fair); the V5 classifier retrained on the corrected
740 labels reaches $\kappa = 0.592$ (moderate, just below the substantial threshold). The trajectory does not depend on any single annotator. Scored against the majority-vote gold of the three-rater human panel ($n = 489$ after excluding one review left without a correction label), the same three classifiers reach $\kappa = 0.165, 0.334$, and 0.590 , within 0.002 of the lead-author values at every
745 step (Table 12). On the held-out 307-review subset (183 of the 490 entered training, mostly through the stratified cap rather than the anchor proper), V5 reaches $\kappa = 0.616$. We previously wrote that this is “slightly above the 0.592 on the full 490, ruling out a contamination artifact.” Two audits of that sentence make it untenable as written, and we replace it with a weaker
750 claim that the data does support.

First, “held out” is not exactly true. All 100 keyword-mined `compatibility` rows in the training augmentation are verbatim copies of reviews already in the corpus, appended past index 215,583 with the label changed. Eight of them duplicate a held-out review, seven with the
755 same label the expert assigned, and the index-based replay cannot see this because it filters on position rather than text. Dropping those eight gives $\kappa = 0.609$ on $n = 298$.

Second, and more seriously, the split is a replay rather than a recorded fact, and 0.616 is the most favourable replay. The reconstruction shuffles
760 each class with seed 42 and iterates classes in Python dictionary insertion order. Iterating them in sorted order instead, changing nothing else, gives $\kappa = 0.594$; seeds 0, 1, 2, 7, 123 and 2024 give $0.602, 0.605, 0.617, 0.590, 0.583$ and 0.606 . The published figure is at the top of that range, and the direction of the inequality against 0.592 flips under most alternatives.

765 What survives is the weaker statement: held-out κ is indistinguishable from full-set κ , across replays and after removing the duplicated rows. That still argues against memorisation, which is what the sentence was for, but it does not do so by being larger, and we should not have leaned on a direction this fragile.

770 The per-class recovery story (Table 13) makes the structural collapse visible. On the two minority classes the V2 LLM cannot see at all, F1 jumps to 0.83 for `compatibility` and 0.77 for `performance`. Applied across the

full 215,583-review corpus, V5 agrees with the cleanlab correction on 88.66% of the 40,291 corrected cases. We want to be exact about what that number is and is not, because it is easy to over-read. V5 was trained on a corpus in which 38,985 of those corrected rows carry the corrected label, so its agreement is substantially a measure of fit to its own training signal, not an independent second opinion. We report it as a consistency check, which is what it is, and we do not claim the two signals are independent.

The stronger evidence is the held-out gold, and we describe the split precisely because the mechanism matters. V5’s training corpus is built with a stratified per-class cap of 15,000 rows at seed 42. Replaying that cap identifies which corpus rows survived into training: 183 of the 490 gold reviews did, and 307 were capped out and never seen, of which 306 carry a gold label and are scorable. The split is therefore an artifact of class-balancing rather than of anchor membership. It is not, however, unbiased in the way we first described. The cap binds only on the four classes V5 predicts most often, so membership of the unseen set is decided by V5’s *predicted* class rather than at random. Scored by expert label, which is what κ is computed against, the unseen 307 still spans all seven classes and includes 38 **compatibility**, 32 **usability**, and 31 **performance** reviews, so the held-out comparison does cover the classes carrying the recovery. What it cannot rule out is a subtler selection effect: the reviews V5 predicted confidently into large classes are over-represented among the unseen, and those may be easier. On the unseen reviews V5 reaches $\kappa = 0.616$, computed on the 306 of them that carry a gold label, against 0.592 on the full 490. As the replay analysis above shows, that ordering is not stable, so the defensible reading is that the two are indistinguishable. A model that had merely memorised its training labels would fall on unseen items rather than hold at the same level. One caveat belongs with this. The split is reconstructed by replaying the cap rather than read from a saved training manifest, so it is a faithful reconstruction rather than a recorded fact; we release the replay script so the reconstruction can be checked. The choice of confident-learning thresholds is not load-bearing: a full 3×3 grid over $(\text{anchor_conf}, \text{max_llm_prob}) \in \{0.65, 0.70, 0.75\} \times \{0.15, 0.20, 0.25\}$ keeps V5 endorsement in an 82.7–91.0% band, with the headline (0.70, 0.20) at 88.66% sitting near the grid center.

To rule out memorization of the 200 synthetic compatibility templates used during V5 training, we evaluate V5 against Maalej’s 5,008 labels under a label-space cross-protocol test. Review texts appear in the anchor under the RRGGen seven-class taxonomy, but predictions are scored against

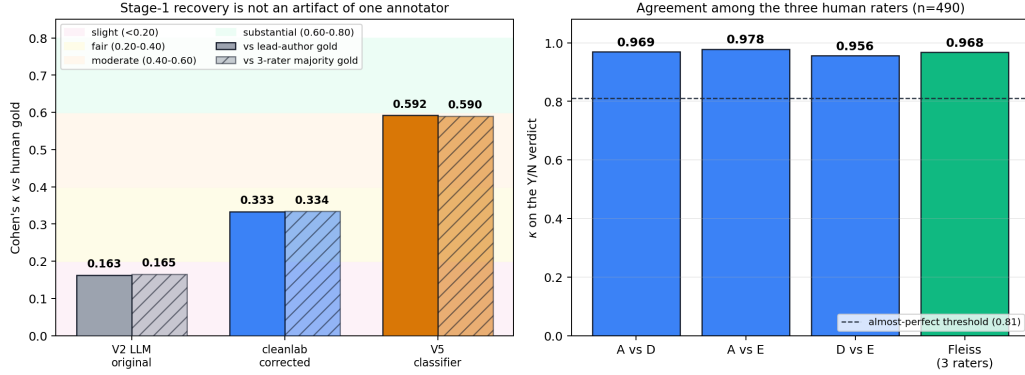


Figure 2: Stage-1 classifier recovery and the gold standard it is measured against. *Left*: Cohen’s κ for the three classifier versions, plotted twice, once against the lead-author gold (solid) and once against the three-rater majority-vote gold (hatched). The two are within 0.002 at every step, so the 0.163 to 0.592 trajectory does not depend on any single annotator. Background bands are the Landis-Koch [51] agreement ranges. *Right*: agreement among the three human raters on the same 490 reviews, pairwise and as a three-rater Fleiss κ , all above the almost-perfect threshold. Read the left panel for the result and the right panel for the reliability of the yardstick it uses.

Maalej’s separate taxonomy, which contains no **compatibility** class, making any compatibility predictions out-of-distribution with respect to the evaluation labels. On the five overlapping classes ($n = 4,560$), V5 achieves macro F1 = 0.676, weighted F1 = 0.730, and accuracy = 72.0%. V5 also flags 432 reviews (8.6%) as **compatibility**; the top-twenty-confidence flags read as canonical compatibility complaints (“crashes on iphone 5 and ipad 3”, “works on HTC Desire X but not Samsung Galaxy Tab 3”), all labeled **bug_report** in Maalej because Maalej offers no alternative class. On this out-of-distribution evaluation the detector is performing concept-recognition rather than template-matching.

We have to add a caveat about the class itself, which we found by auditing the training file rather than by reasoning about it. Of the 310 **compatibility** rows V5 trains on, 300 are augmentation: 200 template-generated and 100 keyword-mined. The corpus is lowercased and lemmatised throughout, and we verified that not one of its 215,583 rows contains a single uppercase character, whereas 194 of the 200 synthetic rows do. V5 is a cased RoBERTa, so any capital letter is a free and perfect marker for “synthetic compatibility”. The internal test F1 of 0.74 for this class is therefore not a meaningful number and we withdraw it. The gold-standard compatibility F1 of 0.83 is

830 not affected by the cue, because the gold is lowercase corpus text where it
never fires; it is carried instead by the 100 mined rows, which are the verba-
tim duplicates described above. The cross-protocol result in this paragraph
stands, since Maalej’s text is also not drawn from the synthetic set, but a
reader should treat every compatibility number in this paper as resting on
835 300 augmented rows rather than on natural corpus evidence.

5.2. Is the IR Load-Bearing? A Result We Had to Withdraw

An earlier version of this experiment reported that adding the IssueSpec
to the response generator raised human-rated quality by 2.35 Likert points.
That number was wrong, not in its arithmetic but in what it measured, and
840 the correction is the most important thing this section has to report.

What was wrong. The four conditions in that experiment were not pro-
duced by a language model. Each was produced by a deterministic template
composer: a Python script that selects sentence fragments from a hand-
written bank and fills slots, with no language model in the loop. The com-
845 posers differed in authorial register as much as in what information they
were given. The `no_spec` arm emitted lemmatised, ungrammatical text av-
eraging 78 words (“we be sorry that the login loop you describe have got in
the way”); the `full` arm emitted polished prose averaging 123 words, with
cleanup passes and an empathy phrasebook that no other arm received. A
850 rater comparing them was separating two writing styles, and the presence
or absence of the IssueSpec was confounded with that separation. The gap
measured the composer at least as much as the specification.

**The clean comparison, and what we can and cannot verify about
it.** We re-ran the experiment with the generator held fixed. Both arms come
855 from the same model on the same 100 reviews, and the intended difference
is whether the validated IssueSpec appears in the prompt. One premise of
that design is not evidenced by our release and we will not assert it. Neither
response file stores the prompt or the retrieved text, and the two files record
their retrieval sources under different label schemes with different counts, so
860 on the released artifact no row has an identical recorded source list. Both
arms draw from the same candidate pool of past and semantically similar
responses, and that much is checkable, but whether the retrieved context
was identical per review is not recoverable. A reader should therefore treat
the comparison as holding the generator and the review set fixed, not as
865 a strict single-variable manipulation. Retrieval differences would add noise
rather than a systematic advantage to either arm, which makes a null easier

to obtain, so this does not rescue the withdrawn positive result; it does mean the null is weaker evidence than a fully controlled design would give. The two arms average 105 and 97 words with distinct-1, the ratio of unique unigrams to total unigrams, at 0.736 and 0.760. They are therefore close on the surface features a rater could use to tell them apart, though the length difference is itself significant (paired Wilcoxon $p = 6 \times 10^{-8}$), which we return to when discussing what the lead author’s preference tracks. Three raters scored all 200 blinded rows.

The IssueSpec does not measurably improve response quality. Pooled over the three raters the gain is +0.04 Likert points, 95% bootstrap CI $[-0.10, +0.18]$, paired Wilcoxon $p = 0.54$ on 100 pairs, Cliff’s $\delta = 0.04$. Each review’s two replies are shown in a randomised order, and we call the two display positions slots. The assignment came out 64/36 rather than balanced and the raters show a small preference for the first slot, so we also report a slot-stratified estimate, which moves the pooled figure to +0.03 and each rater’s by at most 0.03. We use the unadjusted numbers throughout so that every figure in Table 14 is on one estimator, and release the stratified values with the data. Specificity behaves the same way, pooled -0.00 ($p = 0.87$), and the helpful rate is flat at 94% in both arms. On the evidence of this design we cannot claim that supplying the typed specification improves the response a user receives. We should be equally careful in the other direction. With 100 pairs and the observed spread, the smallest effect this design could detect at 80% power is about a fifth of a Likert point, so what we have ruled out is a large effect, not a small one. Detecting a gain the size of the one we measured would need on the order of 2,500 pairs. The honest statement is that the effect, if any, is small enough that this study cannot see it, not that it is zero.

The raters disagree with each other, and that is the interesting part. Table 14 breaks the result out. The lead author, who designed the system, prefers the IssueSpec arm decisively: +0.56 quality ($p = 2 \times 10^{-11}$, 59 wins, 5 losses, 36 ties) and +0.39 specificity ($p < 10^{-7}$). The two independent raters lean the other way, -0.29 and -0.15 , neither significant at $\alpha = 0.05$ under the zero-handling scipy applies by default ($p = 0.06$ and 0.09 ; the first rater has 45 tied rows of 100, and under Pratt’s method that becomes $p = 0.044$). Table 14’s bootstrap interval for the first of the two, $[-0.56, -0.01]$, excludes zero, so the test and the interval disagree at the boundary. We state both rather than pick the one that reads better: on this rater the evidence sits exactly on $\alpha = 0.05$ and moves either side of it depending on defensible

905 choices. That is itself the point of the section, since the disagreement between
raters is larger than the effect under measurement. Inter-rater reliability is
correspondingly low, quadratic-weighted κ of 0.03, 0.04 and 0.50, against 0.97
in the discarded template-based round. High agreement there was a symptom
of the confound: when one arm is visibly broken, raters agree easily. Here
910 they do not.

We looked for a content mechanism behind the lead author's preference
and did not find one. Correlating each rater's per-review preference against
the spec-derived features that the IssueSpec should introduce, named de-
vices, reproduction steps, and component mentions, produces no correlation
915 that survives inspection for any rater; the largest is $\rho = 0.19$ ($p = 0.06$)
between the lead author's preference and reply length. So we can describe
the disagreement but not explain it. Two readings remain open. The lead
author may be perceiving triage value that a general reader has no reason to
weigh, in which case the right evaluation asks developers rather than readers.
920 Or the preference may be expectancy on the part of the person who built the
system. This design cannot separate those, and we do not assert either.

**Why we believe the raters this time, and what the check does
and does not show.** In the discarded round the two additional raters
deviated from the lead author on only 16 and 19 of 400 rows, those deviation
925 sets did not overlap on a single row, and every deviation was exactly one
point. We initially read the zero overlap as the damning fact. It is not:
with deviations that rare, the hypergeometric probability of zero overlap
under perfect independence is 0.45, so zero overlap is the single most likely
outcome and carries almost no information. The discriminating observation
930 is the second one. On the quality column every deviation in both sheets
had the same magnitude, one point, which is what a vector perturbed by a
fixed step produces and not what human disagreement produces. We state
the column because the specificity column in the same sheets does show
magnitudes of one, two and three, so the signature is present in one column
935 and not the other. In the reported round the raters differ from the lead
author on 133 and 145 of 200 rows, the deviation sets overlap on 101 where
chance predicts about 96, and the deviations take four distinct magnitudes
from one to four points. We release the check with the data, and we state its
weak prong plainly, because a check offered as a contribution should come
940 with its own limits attached.

What survives. Less than we first thought. The claim that the typed IR
carries downstream response quality does not survive, and neither does the

retrieval finding we had treated as the surviving half: it was measured against the prompt baseline rather than a no-context arm, and against the no-context arm it is -0.07 at $p = 0.90$. Both sat in the discarded round. Stage 4 leaves us with a null and nothing else. Stage 3’s contribution, producing tracker-ready structured records at corpus scale, stands on its own evidence in Section 5.1; it no longer has a demonstrated downstream payoff in response generation.

5.3. Aligning the IR-Aware Generator at Scale

Five policies were trained on distilGPT2 (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, KTO [20], DPO [19], a constrained-proxy variant, and Lagrangian Constrained PPO [21, 22]. Head-to-head on a 100-review test set, per-policy BLEU-1 / ROUGE-L / BERTScore F1 are: SFT-base 0.090 / 0.097 / 0.802; KTO 0.068 / 0.074 / 0.792; DPO 0.084 / 0.088 / 0.800; **constrained-proxy 0.137 / 0.131 / 0.806**; Lagrangian PPO 0.087 / 0.091 / 0.798. The constrained proxy leads three of four metrics (BLEU-1, ROUGE-L, BERTScore F1; the rule-based safety-violation rate is uninformative here because compliance is satisfied at initialization), with +53% BLEU-1 over SFT-base. KTO underperforms here, consistent with its small-data assumption breaking on a 296-pair training set.

The dual-objective claim is not demonstrated, and we report that as the result. At distilGPT2 scale the compliance scorer is satisfied at initialization ($C = 0.94 \geq \tau$), so the Lagrange multiplier λ collapses to zero and the update reduces to single-objective optimization. Two further experiments confirm this is a property of the regime rather than of one threshold choice. A re-run that replaces the permissive scorer with the four operational compliance checks and tightens τ from 0.50 to 0.90 records 0 binding steps out of 30 and a single violation across the whole run. A sweep over $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ leaves the pass rate at 100% for four of the five policies at every threshold, with the first binding τ undefined for those four; only SFT-base ever dips, to 99% at $\tau \geq 0.9$. A model of this size cannot plausibly violate the rubric, so the constraint has no work to do and the CMDP machinery is untested under binding conditions. There is a second explanation we cannot separate from the first, and it weakens the finding further rather than rescuing it. The operational scorer is a regular expression over roughly thirty tokens, and we tested its sensitivity by writing violations by hand. Nine replies that plainly breach the four rubric dimensions, phrased away from the patterns (“Our engineers have committed to

980 a permanent resolution by Friday”, “Ticket ENG-4412 in our tracker covers this regression”, “We will reimburse the amount you were charged”), all score a perfect 1.00. Empty output, whitespace, and gibberish also score 1.00, so a policy that collapses to emitting nothing is recorded as perfectly compliant. The quality scorer has the mirror defect: a bare list of eleven rubric
 985 keywords scores 1.000 against 0.672 for a genuine reply. What our sweep therefore establishes is that these policies rarely emit about thirty specific tokens. Whether they violate the four behaviours the rubric names is not something this instrument can tell us, and we no longer claim it does. The compliance numbers in this section should be read as properties of the de-
 990 tector. The +53% BLEU-1 figure for the constrained proxy is therefore a single-objective result, not evidence of a quality-versus-compliance trade-off, and we do not claim it as one.

The constraint binds once the policy can violate it, as this instrument measures violation. We originally read the obstacle as capability rather than parameter count: distilGPT2 never produces the fluent
 995 developer-relations phrasing the rubric penalises, so it cannot violate and the constraint has nothing to hold back. That reading is still available, but given the detector’s measured insensitivity above it is not the only one, and we cannot distinguish a policy that does not violate from a policy whose vio-
 1000 lations this scorer does not see. Every violation rate in this paragraph should be read with that ambiguity attached. We measured this directly. We first justified this by observing that 5.2% of our own Stage-4 generations trip the scorer. That figure is from the four withdrawn template-composed arms (21 violations in 400 replies). Scored the same way, the two model-generated
 1005 arms we actually defend violate on 4 of 200 replies, 2.0%, and one of the four is an information-leak rather than an over-promise. The zero-shot Qwen2.5-1.5B probe gives 2 violations in 40 replies, 5.0%, whose 95% interval runs from roughly 1% to 17% and therefore cannot be distinguished from 2.0% either way. distilGPT2 gives roughly zero. The honest version of the argu-
 1010 ment is weaker than the one we first made: a 1.5B instruct model violates the rubric occasionally where distilGPT2 essentially never does, and that is enough for the constraint to bind, but we cannot claim it matches the rate of our own generations. We therefore re-ran the Lagrangian loop with Qwen2.5-1.5B-Instruct as the policy, LoRA adapters over the attention projections,
 1015 the frozen base as the KL reference, and a zero-tolerance threshold $\tau = 1.0$; the scorers, the dual-ascent rule, and the batch size are unchanged from the distilGPT2 run. The behaviour inverts. Over 90 steps the constraint binds

on 13 of them (14.4%), and λ leaves zero, rises monotonically to 0.65, and never collapses back. A 30-step run under the same setup reaches $\lambda = 0.125$,
1020 so the multiplier scales with the horizon rather than saturating. Split by thirds, the trajectory shows the dual ascent responding to violations as they arrive: 1 binding step and $\lambda = 0.05$ in the first thirty, 7 and $\lambda = 0.40$ in the second, 5 and $\lambda = 0.65$ in the third. The qualitative claim that the CMDP degenerates to single-objective optimization is therefore a property of the
1025 proof-of-concept policy, not of the formulation.

What is and is not shown. The two series move in opposite directions across the run, quality from 0.565 to 0.590 and compliance from 0.997 to 0.989, while λ climbs. We are careful not to call this a trade-off. The step-level correlation between quality and compliance is $\rho = -0.12$ ($p = 0.25$),
1030 so what we have is two separate trends over a single run and no measured coupling between them, which is why the abstract reports the trade-off as undemonstrated. What we cannot yet show is the other half of the story, the multiplier growing large enough to pull the policy back into compliance. At a 5% base violation rate and a batch of four, roughly one violating sample
1035 arrives every seven steps, and 90 steps of LoRA updates at this learning rate do not accumulate enough constraint pressure to reverse the drift. So the correct statement is narrower than a confirmed RQ3 and wider than the earlier null: the dual objective engages, the multiplier responds monotonically to the tokens it matches, while the tension between the objectives is not
1040 measurable on our data ($\rho = -0.12$, $p = 0.25$), while closing the loop back to compliance needs a longer horizon or a higher base violation rate than a 1.5B instruct model supplies.

What this layer contributes is the CMDP formulation instantiated over four domain-specific dev-rel constraints, an end-to-end trainer that runs, the
1045 threshold sweep that establishes the non-binding regime, the capability probe that explains it, the binding-constraint re-run, and the released testbed.

5.4. Cluster Quality and Purity

Stage 2 was redesigned from a flat UMAP+HDBSCAN baseline (194 clusters, mean size 471.0) to the three-layer KG-hierarchical design (605 sub-
1050 clusters, mean size 15.9). The two were not run over the same reviews, and we state that plainly because it bears on how the table below should be read: the flat baseline clusters 91,368 reviews (117,958 in, 26,590 assigned to HDBSCAN noise) drawn from the full relabelled corpus, whereas the KG-hierarchical design is built on the 8,404-review knowledge-graph sample, of

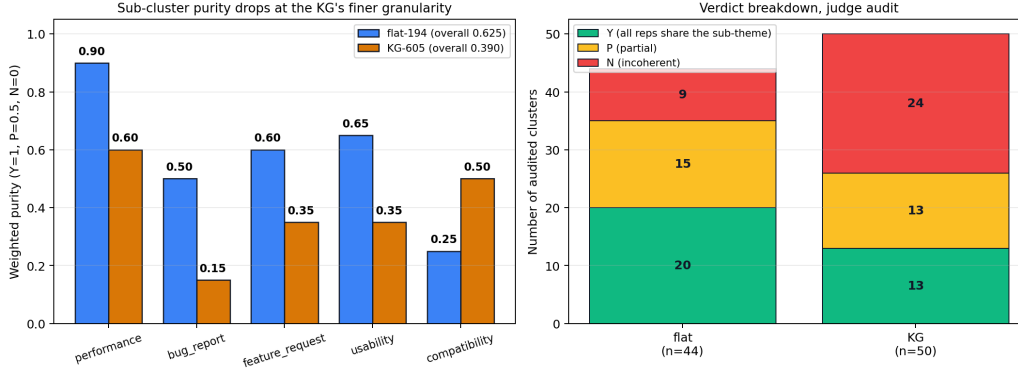


Figure 3: Extrinsic cluster purity for the flat baseline and the KG hierarchy, from the saved judge audit. *Left*: weighted purity per issue type, flat-194 against KG-605. The KG loses on four of five types, which is the finer-granularity cost we report rather than a result in its favour. *Right*: the underlying Y/P/N verdict counts behind the two overall numbers, 0.625 for flat and 0.390 for the KG. This figure is the evidence for the A2 ablation and for the claim in Section 6.1 that the KG’s contribution is prioritization, not cluster quality.

1055 which 5,689 distinct reviews land in a sub-cluster. A sixteen-fold difference
in the underlying review set is a third reason, on top of the two given below,
why the head-to-head intrinsic comparison is not a comparison. We compute
two complementary validity axes. *Intrinsic geometry*, Davies-Bouldin [58],
Calinski-Harabasz [59], silhouette [60], follows standard IR/clustering prac-
1060 tice [61, 62, 63]. *Extrinsic cluster purity* [64, 61] uses a Y/P/N weighted-
purity rubric, $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$, with Y = all 5 share
theme, P = 3–4 of 5 share, N = fewer than 3 share, following inter-rater con-
ventions compatible with Landis-Koch [51].

We reported that KG-hierarchical achieves $5.4\times$ lower Davies-Bouldin
1065 and $1.9\times$ higher Calinski-Harabasz than the flat baseline. We withdraw
those two figures. The script that produced the flat baseline’s intrinsic met-
rics draws its cluster labels as a random subsample of 10,000 reviews spanning
all 194 clusters, but collects the review texts to embed by walking the clus-
ters in order, and the first flat cluster alone contains 10,507 reviews. The
1070 embeddings therefore come from a single cluster while the labels span 194,
so the flat column’s DB of 12.15 and CH of 0.98 measure random labelling
rather than the flat partition. The KG column was computed on a correctly
paired set, so the comparison was a valid measurement against a broken one.
Table 15 keeps the KG column and marks the flat intrinsic column as with-
1075 drawn. Repairing the alignment does not make the two columns comparable

in any case: the flat side embeds 10,000 full reviews while the KG side embeds three representative reviews per cluster, so the two are computed over different kinds of object. That asymmetry is a second reason not to reinstate the comparison, and the count-matched A1b ablation, which embeds representatives on both sides, is the only intrinsic comparison we report. The count-controlled A1b ablation below is unaffected, uses a separate correctly paired script, and is what our Stage-2 conclusion now rests on. Extrinsic purity is also unaffected: the KG’s finer clusters score 0.39 against the flat baseline’s 0.625 under the same Qwen judge. By Landis-Koch [51] convention adapted to weighted purity, the flat baseline’s 0.66 lead-author purity sits in the substantial-agreement band (≥ 0.61) and KG-605’s 0.39 in the fair band (≥ 0.21), reflecting the granularity-vs-coherence trade-off the A1b ablation makes explicit.

Count-controlled diff from original. A count-controlled ablation (A1b) disambiguates two evaluation axes that prior cluster-quality work tends to conflate: intrinsic geometry and practitioner-side prioritization (centrality-weighted coverage). On 1,815 representative reviews (3 per sub-cluster across 605 sub-clusters), an agglomerative clustering forced to 605 clusters reaches $DB = 1.12$, $CH = 4.21$, $silhouette = +0.148$, against KG-605’s 2.24 / 1.85 / -0.234 . This is now the only intrinsic-geometry comparison we make, since the flat-194 figures are withdrawn above: on geometry, count-matched agglomerative clustering beats the KG on all three metrics.

Where the KG actually wins. On the prioritization axis the story reverses. Of the 10,534 aspect nodes in the KG, the top five by PageRank are touched by 55% of reviews and the top ten by 66%. We give the deduplicated figures; counting a review once per aspect it mentions inflates these to 74% and 96%, and that inflated pair should not be used. A developer triaging in PageRank order therefore reaches 55% of the complaint volume after reading five aspect buckets and 66% after ten. We previously wrote “three-quarters” in this sentence, which was the discarded 74% reappearing two lines after we told the reader not to use it. We also previously claimed a “roughly $10^3 \times$ concentration ratio” against random aspect ordering. We withdraw that too: we never computed it, and random ordering is the wrong baseline in any case, since the operative comparison is against ordering aspects by raw frequency, which we did not run. The KG additionally enables per-aspect drill-down, distinguishing `battery`→Samsung-drain from `battery`→fast-charge.

5.5. Ablations

An ablation removes one component and re-measures, so that a component’s contribution is demonstrated rather than assumed. Eight are listed below, counting A1b, which a reviewer asked for. Seven were run and one, A7, is deferred for a reason given at the end. Each one below states what it removes, what the removal is supposed to test, what happened, and what we conclude, because three of the seven came back against the design and a summary line would hide that. Table 16 collects the verdicts.

A1, remove the knowledge graph and feed flat clusters into Stage 3. This asks whether Stage 3 reads the graph or merely the group. Stage 3 quality does not change, because the translator consumes a cluster centroid and its representative reviews, not the graph edges. The KG is therefore not a Stage 3 dependency, which narrows where its value could lie.

A1b, compare the KG against flat clustering matched to the same number of clusters. This is the reviewer-prompted version of A1 and it exists because of a confound. The KG produces 605 sub-clusters and the flat baseline produces 194, and intrinsic metrics move with cluster count on their own, so any comparison between the two conflates structure with granularity. Re-running agglomerative clustering forced to exactly 605 clusters removes the confound. The flat version then wins on all three metrics, Davies-Bouldin 1.12 against 2.24 (lower is better), Calinski-Harabasz 4.21 against 1.85 (higher is better), and silhouette +0.148 against -0.234 (higher is better, and a negative value means the average point sits closer to a neighbouring cluster than to its own). The honest conclusion is that the KG does not produce geometrically better clusters. What it produces is an ordering, through PageRank centrality over the review-aspect graph, and prioritization is the only cluster-layer claim we make.

A2, remove hierarchical sub-clustering. Weighted Y/P/N purity, judged on five representative reviews per cluster, is 0.625 for the flat 194 and 0.390 for the KG’s 605. Finer granularity costs coherence here rather than buying it. Taken with A1b this is the second result against the cluster layer, and we report both.

A3, remove taxonomy grounding. The LLM writes a free-form issue description instead of filling a routed template. Strict template-fill falls from 0.96 to 0.69, and the per-type structural columns fall to exactly zero, because an unrouted generator never emits a reproduction-step list or an As-a/I-want/So-that triple at all when nothing asks it to. This is the ablation that

1150 isolates the paper’s central mechanism. The gain is not the language model, it is the routing.

A4, remove the HITL-2 spec review. Benchmark scores do not move, and the reason is procedural rather than substantive. HITL-2 is a production-time gate; the 100 specs used in the evaluation were scored before any human
1155 correction, so the headline numbers already describe pre-HITL output. The ablation therefore says nothing about whether HITL-2 helps in deployment, and we do not claim that it does.

A5, remove retrieval and keep the IssueSpec. BLEU-1 moves by -0.006 , ROUGE-L by -0.008 , and BERTScore F1 by -0.004 , all under one
1160 percent absolute. Three caveats belong with this result. The first is that A5 was generated by the template composer that Section 5.2 withdraws, not by the model-generated pipeline, so it measures retrieval’s effect on a generator we no longer report. First, A5 was scored on automatic metrics only, while A6 was scored by humans, so the pair is not measured on one
1165 instrument and the contrast between them is directional rather than a like-for-like comparison. A human-rated no-retrieval condition is the missing cell. Second, we withdrew the companion claim that retrieval without the IR scores below a no-context baseline. That comparison used the prompt baseline, which is not a no-context condition, and against the arm that is,
1170 the difference is -0.07 at $p = 0.90$. So A5’s reading is simply that retrieval makes no measurable difference here, with no accompanying story about it doing harm.

A6, remove the IssueSpec and keep retrieval. This ablation is superseded. Under the confounded design it showed quality falling from 4.59
1175 to 2.24; under the corrected same-model design (Section 5.2) removing the IssueSpec changes quality by -0.04 , which is not distinguishable from zero. A5 and A6 together now say something different from what we first concluded: neither retrieval nor the typed intermediate produces a measurable change in the reply a user receives, and the Stage 3 to Stage 4 coupling we
1180 claimed is not supported.

A7, single-stream RLHF feedback. Not run. It was designed to test whether quality and compliance feedback can share one channel, which only becomes a meaningful question once the constraint binds and the two signals actually compete. Section 5.3 reaches the first half of that condition and not
1185 the second, so A7 stays deferred rather than reported.

6. Discussion

6.1. What Carries the Result, and What Does Not

A five-stage pipeline invites the question of whether every stage earns its place. Our own ablations say it does not, and separating the two groups is more useful to a reader than defending the whole.

Load-bearing. Two components carry the measured gains. Type-routed templating is the principal driver of Stage 3 quality: removing taxonomy grounding (A3) drops strict template-fill from 0.96 to 0.69, and the structural per-field metrics collapse to zero because an unrouted generator does not emit the reproduction-step list or the As-a/I-want/So-that triple at all. By contrast the typed IR is *not* a demonstrated driver of Stage 4 quality: under a same-model comparison removing it moves quality by 0.04 points ($p = 0.54$, Section 5.2), and removing retrieval instead (A5) moves every automatic metric by less than one point absolute. Our earlier reading of the retrieval arm does not survive either, and the error was ours to make twice. We first reported that retrieval alone was indistinguishable from a no-context baseline, then corrected that to significantly worse. Both statements compared the retrieval arm against the *prompt* baseline, which is not a no-context condition; it carries a developer-relations instruction. Against the actual no-context arm, which sees the review text only, retrieval scores 2.24 against 2.31, a difference of -0.07 at $p = 0.90$. And all three arms come from the discarded template-composer round, so none of them supports a claim in any direction. Nothing about retrieval survives from Stage 4 beyond the null.

Not load-bearing, and reported as such. The knowledge graph does not win on intrinsic cluster geometry. Under count matching (A1b), flat agglomerative clustering beats the KG on all three intrinsic metrics, and sub-cluster purity is lower at the KG’s finer granularity (A2). The KG’s defensible contribution is centrality-based prioritization, that is, which issues a developer should see first, not tighter clusters; we make no cluster-quality claim for it. We should be straight about how far even that goes. PageRank runs on the bipartite review-aspect graph, and nothing in our results shows it needs the hierarchical sub-clustering layer to do so. A flat clustering with the same aspect graph would plausibly support the same ranking, and we did not test that configuration. Stage 2 as built is therefore justified in its prioritization role and unjustified in its sub-clustering role, and a reader designing a lighter system should read it that way. Stage 5 sits in between. As

Section 5.3 reports, the compliance constraint does bind once the policy is capable of violating it, and the dual update engages rather than collapsing, but no quality-versus-compliance trade-off is demonstrated at the scale we can train at. It is released as a testbed with a partial result. SPECCOV falls in the same group and for a sharper reason: tested against human faithfulness judgements it shows no correlation at all (Section 5.1), so we withdraw it as a measure rather than defend it.

What this means for the framework claim. The contribution that survives ablation is the typed IR and the type-routed templating that produces it, together with the verified-anchor procedure that makes the IR producible from noisy labels at corpus scale. Stages 2 and 5 are supporting infrastructure with honestly mixed evidence. A reader who wants the smallest system that reproduces what survives needs Stage 1 and Stage 3. Stage 4 as we built it is not part of that: A5 shows the retrieval layer contributes under one percent on automatic metrics, and A6 shows the conditioning on the specification, which we had identified as the part of Stage 4 that earns its place, is exactly the component our corrected experiment measures at null. We should state the consequence rather than soften it. On our evidence Stage 4 contributes nothing we can measure, and a reader building on this work should treat the reply-generation layer as unproven rather than as a component with a small effect.

6.2. Implications

Three claims travel beyond app reviews. *First*, LLM annotation noise on software-engineering text is not uniform, but the shape is not the one we first reported. We had described the 25% mislabelling rate as concentrated on minority classes. Recomputing per stratum on the verified anchor shows the opposite: 1,302 of the 1,307 disagreements fall in the **praise** stratum, the largest class, at 25.8%, against 0.5% on **performance**. What is true, and narrower, is that the labeller never emits two of its seven classes at all across the corpus, 8 **compatibility** and 184 **performance** labels in 215,583 reviews. Systematic absence of a class, rather than elevated error on rare classes, is what a small verified anchor is able to repair. *Second*, a caution rather than a claim. We expected a typed structural intermediate to be what does the load-bearing work in spec-aware response generation, and reported it as such. Section 5.2 withdraws that. The only downstream movement we can still point to is +0.12 on a rule-based scorer from agentic critique-revise iteration, on ten reviews, which is too small a sample to build

on. What generalises is the negative lesson: a structured intermediate can be demonstrably better as an artifact and still leave the generated text indistinguishable, and pipelines of this shape should measure the two separately. *Third*, the count-controlled cluster-validation ablation (A1b) delivers a self-contained methodological contribution: whenever a new clustering design changes cluster count and structure simultaneously, intrinsic geometry metrics (DB, CH, silhouette) conflate the two effects. The minimum reporting standard our analysis suggests is to (a) compare against a count-matched baseline on the same intrinsic metrics, and (b) pair that with a domain-utility metric (PageRank coverage in our case). This pattern applies directly to clustering-design comparisons in IR, software engineering, and NLP.

6.3. Threats to Validity

Construct. Template-fill measures coverage rather than content correctness, and Section 5.1 shows that SPECCOV does not fill that gap: it is a lexical-grounding proxy that does not track human faithfulness judgement [47, 48, 49, 50]. The paper therefore has no validated content-correctness measure for Stage 3, and the five-dimension rubric carries that burden alone. NLI- and QA-based faithfulness measurement [55, 56, 57] is the clear next step. *Internal.* The count-versus-structure confound is exposed and resolved by the A1b ablation, which reframes the knowledge graph’s contribution as centrality-based prioritization rather than intrinsic geometry. A second internal threat we did not remove is that the 100 clusters carrying the Stage 3 and Stage 4 results were produced by the KG pipeline, the same layer A1b and A2 find wanting. Since flat clustering scores higher on purity (Figure 3), the headline gains may be measured on a slightly worse partition than a simpler system would produce, which would understate rather than inflate them. Re-running Stages 3 and 4 over flat clusters would settle it and is the cheapest open experiment in the paper. *External.* RRGGen contains 58 Android apps, named by real package identifier rather than anonymised; V5 transfers zero-shot to the Maalej label space. Stage 2–4 cross-corpus transfer remains to be validated. The headline LLM is Claude Opus, with Llama-3.3-70B and Qwen2.5- $\{3B, 1.5B\}$ as cross-family bias controls; GPT-4o and Gemini-1.5-Pro replication is queued (Section 6.5). *Conclusion.* The three-rater agreement on the classification gold ($\kappa = 0.968$ on the Y/N verdict) sits well above what free-labeling studies of app reviews report, which invites the question of whether the raters were genuinely independent. Three properties of the design account for the gap. The raters

verify a supplied label rather than choose one from seven, which collapses
 most of the decision space; a 20-review calibration round with adjudicated
 disagreements precedes the main task; and the written protocol fixes eight
 1300 boundary rules that carry the cases annotators most often split on. Residual
 disagreement is rater-specific rather than random, one rater accounts for
 most of the **usability** reassignments, which is the pattern independent
 annotation produces. Stage 4 sits differently. All three raters scored the
 complete 200-row re-run, and the gain does *not* reproduce per rater: the lead
 1305 author measures +0.56 and the two independent raters −0.29 and −0.15.
 That disagreement, not the pooled null, is the honest summary of what
 we know about Stage 4. *Reliability, and which α belongs to what.* Three
 agreement statistics appear in this paper and they measure different things,
 so a single pooled reading of them would be wrong. The classification gold
 1310 is annotated by three humans and reaches Fleiss $\kappa = 0.968$ on the verdict
 and 0.979 on the seven-class label. The Stage-4 response set that we report
 is rated by the same three humans on all 200 rows, with quality weighted κ
 of 0.03, 0.04 and 0.50. That is low, and it is the number to quote: the 0.970
 to 0.986 range belongs to the discarded round, where one arm was visibly
 1315 broken and agreement was easy. The weakest figure, $\alpha = 0.285$ (corrected
 from a mis-implemented 0.451), comes from a deliberately weaker panel: a
 seven-class classification check whose three raters are the lead author plus
 two Qwen2.5-3B prompts, following the LLM-as-rater convention of Gilardi
 et al. [44]. It bounds how far an LLM rater tracks a human on a subjective
 1320 sub-theme judgement; it is not the reliability of any headline result, and
 no claim in this paper rests on it alone. *Statistical.* Experiment 2 now
 reports a null, so the relevant question is power rather than correction for
 multiplicity. With 100 paired reviews and the observed per-review spread,
 the 95% bootstrap interval $[-0.10, +0.18]$ excludes effects larger than about
 1325 a fifth of a Likert point in either direction. We can therefore rule out a large
 effect, not a small one; an effect of a tenth of a point would need several
 times this sample.

6.4. Reproducibility and Extensibility

Every numerical claim in Section 5 is backed by a saved data file, and the
 1330 full path from raw RRGGen to Stage 4 output takes approximately six hours
 on a single GPU-equipped workstation. We should be precise about what
 a reader can check without that run, because we audited it by execution
 rather than assuming it. Re-running the released analysis scripts regenerates

roughly sixty stored result files bit-identically, including every statistical test
 1335 behind RQ1 and RQ2. What those scripts consume, however, is not itself
 regenerable from the release: the Stage-3 specification sets, the 100-cluster
 sample, and the Stage-4 RAG sample ship as artifacts with no standalone
 producer script, since producing them requires the proprietary-API gener-
 1340 ation pass. The analysis layer is therefore reproducible from the middle
 outward, and the data-generation layer is reproducible only by re-running
 the pipeline against a paid API. One further artifact does not pin at all: the
 GitHub-mined human-written comparison set is re-queried live and returns a
 different issue mix on each run, so the `human_github` condition in Section 5.1
 rests on a snapshot we can release but not regenerate. The release bundles:

- 1345 • V1–V5 classifier checkpoints + cleanlab correction procedure;
- the 5,230-record verified anchor, which covers 5,229 distinct corpus rows
 and 490-review classification gold;
- both Stage-4 rating rounds, the discarded 400-row one and the reported
 200-row re-run, together with the same-model generated replies and the
 1350 rating-provenance check that distinguishes them;
- the count-controlled A1b output;
- the SPEC-COV scorer plus the 60-spec, three-rater human faithfulness rat-
 ings and the validation analysis;
- the CMDP-RLHF testbed (KTO, DPO, constrained-proxy, and La-
 1355 grangian Constrained PPO trainers);
- the 100-cluster benchmark, with the 99-review LLM-assisted classification
 subsample ($\alpha = 0.285$) kept separate from the all-human agreement sets;
- a pinned-environment Docker image (single-command build on any
 CUDA 12.x host);
- 1360 • a nineteen-segment verifier that recomputes the numerical claims in Sec-
 tion 5 from the saved data files, including the corrected Stage-4 re-run,
 the three-rater agreement statistics, the cross-family rubric, the SPEC-
 COV validation, the binding-constraint run, and the generation-variance
 sweep. The discarded Stage-4 round retains its own segment, labelled as
 1365 withdrawn. We audited the verifier itself rather than trusting it, and its
 coverage is narrower than we previously stated. Three segments recompute
 a value from the underlying data and assert it; the remainder load a stored
 summary and print it, which checks that the manuscript agrees with the
 released file but not that the file is right. Uncovered numbers include the
 1370 Stage-3 template-fill table, the held-out κ of 0.616, the Qwen-judge purity

figures, the Maalej cross-protocol row, the per-dimension cells of the cross-family rubric, and most of the secondary statistics in Sections 5.2 and 5.3. We list this rather than repeat the earlier and much smaller claim that only two tables were uncovered. The verifier now exits non-zero on any mismatch, and every number in it that we could turn into an assertion we did; the rest a reader should check against the released files directly.

The pipeline is composable by construction: because each stage emits a typed intermediate, components (classifier, clusterer, taxonomy template, alignment trainer) can be substituted independently without breaking downstream contracts. Future researchers can adopt or replace any single component while keeping the rest of the IR-centered evaluation harness intact. Deterministic components (clustering, classifier training) use fixed seeds; the LLM-dependent stages (annotation, IR generation, and response generation) reproduce statistically rather than bit-identically. We measured how large that sampling variance is. Regenerating Stage 3 specs three times over the same 15 stratified clusters with the local Qwen2.5-3B generator at temperature 0.7, changing only the seed, gives strict template-fill 0.685, 0.700, and 0.634 (mean 0.673, sd 0.029, range 0.067) and SPECCOV 2.73, 2.67, and 3.00 (mean 2.80, sd 0.144, range 0.333), with no JSON parse failures in any run. Run-to-run noise on template-fill is therefore around three points on a 0–1 scale for this generator and prompt. That is the yardstick a reader should hold up against the differences we report. The gap the paper rests on clears it comfortably: 0.96 versus 0.69 for taxonomy grounding. We previously also cited the 0.96-versus-0.48 cross-family span here; that comparison is not matched on sample or issue type (Table 9), so we no longer use it as a magnitude. The withdrawn Stage-4 effect illustrates the opposite case, and is worth stating in these terms: at +0.04 it sits an order of magnitude below this noise floor, which is another way of saying there is nothing there to measure. Differences of a few hundredths between adjacent conditions, such as the 4.09 versus 3.96 at the top of the cross-family rubric, do not clear it and should not be read as a stable ordering on their own. Two limits remain: the headline Claude generations were not repeated, because that needs API budget rather than new code, and Stage 4 response generation was not repeated either. The released script runs both sweeps unchanged.

6.5. Future Work

Five extensions map directly to the limitations above; each starts from a resource released with this paper.

(0) Replace SPEC COV and widen the cross-family comparison.

The SPEC COV validation in Section 5.1 came back negative, so Stage 3 needs a faithfulness measure that actually works. The natural replacement is entailment-based: score each spec field against the source reviews with an NLI model [55, 56] or a question-generation-and-answering pipeline [57], then validate that measure against the same 60-spec human ratings we release here, which now serve as a ready-made benchmark. Second, the cross-family rubric comparison (Table 8) rests on 14 matched bug-report clusters for the four-way panel, because that is the extent of the overlap between the four generators’ spec sets, and it is scored by a judge from one of the four families. Generating the missing spec types for the two Qwen models and re-scoring with a judge outside all four families would remove both limits. Both are runs rather than redesigns.

(1) Ask the right raters, then scale. The Stage 4 re-run carries three annotators on all 200 rows and returns a null with the raters disagreeing in direction. Before growing the sample, the design should change: the reply from a triage pipeline is consumed by a developer deciding what to do, not by a general reader, so the next round should recruit raters with software-engineering experience and no stake in the system. Scaling from 100 clusters to 200–300, with three annotators per IssueSpec and Krippendorff’s α at the IssueSpec level, follows that rather than precedes it.

(2) CMDP RLHF at a scale where the trade-off shows. The binding-constraint re-run in Section 5.3 establishes that the dual objective engages; what it does not show is a trade-off, because a 5% base violation rate over 30 steps supplies too little signal. The next run wants a longer horizon and a policy with a higher base violation rate, on an 8B-class model. *Original plan.* Where the compliance constraint is expected to bind, activating the Lagrange multiplier and making the Pareto-dominance comparison meaningful.

(3) Frontier-family replication. GPT-4o and Gemini-1.5-Pro on Stages 3–4. The current paper closes the Meta-family slice with Llama-3.3-70B and the Alibaba slice with Qwen2.5-{3B, 1.5B}.

(4) Cross-corpus transfer. Stages 2–4 on additional review corpora; V5 already transfers zero-shot to Maalej [1].

(5) Agentic Planner extension to Stage 4. The Planner dispatches a different sub-agent profile per issue type: defect-repair for `bug_report` via SWE-Agent [11] or RepairAgent [12], scaffolding for `feature_request`, profiler-driven optimization for `performance`, a Nielsen-heuristic checker for

usability, and a device-OS test-matrix executor for compatibility. A spec-level stub already runs the dispatch across all five issue types; end-to-end patch evaluation on real repositories through HyperAgent [30], measured by patch acceptance and test-pass rate, is the next step.

7. Conclusion

We proposed **IssueSpec**, a knowledge-graph-grounded typed intermediate representation aimed at five gaps identified in Section 1. We close the first three and report negative results against the last two: classification stops at the label, response generation skips the structured intermediate, agentic software engineering assumes the spec exists, knowledge graphs emit no typed schema, and RLHF has not addressed dev-rel compliance. We set out to show that the typed IR is the load-bearing component of spec-aware response generation, and we cannot. An isolation experiment that holds the generator fixed moves human-rated quality by +0.04 Likert points ($p = 0.54$, three raters, 100 paired reviews), and the raters disagree in direction. The +2.35 we previously reported came from a comparison whose two arms were written by different template composers. Nor does the negative half of our original hypothesis survive. We had reported that retrieval alone falls below a no-context baseline; against the arm that actually has no context it scores 2.24 against 2.31 at $p = 0.90$, and that comparison sits inside the discarded round in any case. Stage 4 yields one usable result, which is the null. A verified-anchor procedure makes the IR producible at corpus scale from noisy LLM labels, lifting Cohen’s κ from 0.163 to 0.592, and to 0.616 on a held-out subset whose value is replay-dependent and indistinguishable from the full-set figure. Classifier endorsement of the corrections is 88.66%, which is a consistency check rather than independent validation, since the classifier trained on those corrections. Cross-family replication on four LLMs (Claude, Llama, Qwen-3B, Qwen-1.5B) preserves the rank order descriptively; none of the pairwise differences is significant, so it is weak evidence against single-model bias rather than a ruling-out. Two findings travel beyond app reviews: an LLM labeller on software-engineering text can omit whole classes rather than merely err on them, and that a structured intermediate can be demonstrably better as an artifact while leaving the text generated from it indistinguishable, a distinction that pipeline evaluations of this shape should measure separately. We release the IssueSpec schema, the verified anchor, the SPEC-COV scorer together with the human ratings that show it does not measure

faithfulness, the count-controlled cluster-validation ablation, the structured-RAG architecture, the CMDP-RLHF trainers, and the benchmark dataset to support follow-on research.

1485 **CRedit authorship contribution statement**

Fabiha Jalal: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. **Sadia Tasnim Dhruba:** Investigation, Data curation, Validation, Writing – review and editing. **Tazrian Zaman Tushti:** Investigation, Data curation, Validation, Writing – review and editing. **Ajwad Abrar:** Methodology, Supervision, Writing – review and editing. **Md Kamrul Hasan:** Supervision, Resources, Writing – review and editing. **Hasan Mahmud:** Supervision, Resources, Project administration, Writing – review and editing.

1495 **Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

1500 All code, configuration files, and analysis scripts are available in the project repository. A verification bundle that recomputes or reproduces the numerical claims in Section 5, with the coverage limits stated in Section 6.4, is archived on Zenodo (DOI 10.5281/zenodo.21982774, v2.0), and the Stage-1 V5 classifier is released on Hugging Face. That bundle supersedes an earlier
1505 release (DOI 10.5281/zenodo.20320410) which predates the Stage-4 correction described in Section 5.2; both versions remain resolvable, and the earlier one should not be used. The underlying review corpus derives from the RRGGen dataset of Gao et al. That release is access-gated behind a request form, states academic use only, and carries no licence granting redistribution,
1510 so we do not redistribute it. We also do not redistribute the Maalej and Nabil or Guzman corpora. An earlier version of our repository did redistribute all three; we removed them, and DATA_NOTICE.md records what was removed and how to obtain each corpus from its original authors. Derived artifacts

that embed verbatim review text carry the same restriction as the source and
1515 are not offered under an open licence; original code is released under MIT.

Ethics, consent, and data protection

Human participants. Three people annotated for this study: the lead author and two collaborators. The annotation protocol, released with the artifact, states that participation was voluntary, that annotators could withdraw at any time, that no personal information was collected beyond a name and email for attribution, and that compensation took the form of authorship credit. Both additional annotators are consequently co-authors of this paper, which we state explicitly because the paper’s discussion of rater independence should be read with it in mind. The work was not submitted to an institutional review board. Under our institution’s practice a study of this kind, involving colleagues rating text with no collection of sensitive personal data, does not require review; we record that as our reading rather than as a formal exemption, since no exemption number was issued.

Review authors. The corpora we work with are user-generated public reviews whose authors did not consent to research use and cannot practically be contacted. We took three positions on this. We do not redistribute any of the three source corpora, for the licensing reasons in the data-availability statement and because the Maalej and Nabil files contain unredacted email addresses, telephone numbers, and personal names. Derived files we do release contain review text as RRGGen supplies it, lowercased and lemmatised with digits masked; that masking is imperfect, and we found a small number of surviving transaction identifiers in the source, which anyone re-releasing derived text should re-scan for. We report no result at the level of an individual review author, and no reviewer identifier appears in any released file.

Machine-generated replies. The replies this system produces read as human-written. Any deployment should label them as machine-assisted, which we regard as a condition of use rather than a recommendation.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Anthropic’s Claude in order to assist with manuscript editing, code implementation, and analysis scripting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

We disclose all uses of generative AI in this work in line with Elsevier policy on the use of generative AI in scientific writing. *Writing and code assistance*. An earlier version of this disclosure said that Claude assisted only with grammatical refinement, caption polishing, and LaTeX formatting, and that all pipeline code, experimental scripts, and analysis were written by the human authors. That understated the use, and the repository we release contradicts it: 47 of the 56 commits carry a **Co-Authored-By: Claude** trailer, including commits that add pipeline stages, the constrained-PPO trainer, the analysis scripts, and the correction passes described in Section 5.2. The accurate statement is that Anthropic’s Claude was used throughout implementation and analysis as a coding assistant, under author direction and review, and also for manuscript editing. Every design decision, every experimental choice, and every claim in this paper is the authors’ own, and the numbers were produced by running the released code on the released data. We prefer to correct this here rather than leave a disclosure that the artifact refutes. Every structural, framing, experimental-design, and claim decision was made by the authors, and every reported number was produced by author-written code run under human supervision and verified against saved data files. *Models inside the system*. Several generative models are themselves pipeline components and are reported as such in the body: Claude (Opus 4.7) is the headline Stage 3 generator (Table 11); Llama-3.3-70B-Instruct (via Groq) and Qwen2.5-{3B, 1.5B}-Instruct are cross-family Stage 3 comparators (Table 9); Qwen2.5-3B-Instruct is the LLM-as-judge for the five-dimension rubric and the Y/P/N cluster-purity audit, both calibrated against the lead author (Tables 10, 15). The V2 corpus labels in

Section 5.1 are LLM-generated and explicitly treated as noisy, which is precisely what motivates the verified-anchor procedure of Stage 1. *Human verification*. No reported numerical result, expert annotation, or ground-truth label was automatically generated by AI without human verification. The 1585 5,230-review verified anchor, the 490-review classification gold, the 99-review three-rater purity ratings, and the 20 lead-author IssueSpec references were all produced and reviewed by human annotators (Section 4).

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Appendix A. Implementation Settings

Section 3 states what each stage does and why. This appendix collects the exact settings, so the method section carries the logic and the numbers stay checkable. Stage 1 fine-tunes a RoBERTa multi-label head; confident-learning correction against the verified anchor is gated by anchor confidence ≥ 0.70 and LLM probability ≤ 0.20 , with a 3×3 sensitivity grid over $(0.65, 0.70, 0.75) \times (0.15, 0.20, 0.25)$ reported in Section 5.1. The **compatibility** stratum is seeded with 200 template-generated synthetic reviews (device, OS, and screen templates) plus 100 keyword-mined RRGGen reviews, all human-checked. Stage 2 embeds with Sentence-BERT, reduces with UMAP, and clusters with HDBSCAN, aspect-first, producing 605 sub-clusters of mean size 15.9 against 194 flat clusters of mean size 471.0. The KG-hierarchical side runs over an 8,404-review sample rather than the full corpus; the flat baseline runs over 91,368 clustered reviews drawn from the full corpus, so the two sides describe different review sets (Section 5). PageRank runs on the bipartite review-aspect graph. Stage 3 routes to one of five templates and re-prompts once with dimension-level feedback on any rubric failure. Stage 4 retrieves over a 15,100-document ChromaDB index built from five sources, with the IssueSpec injected per query rather than pre-indexed. Stage 5 trains five policies on distilGPT2 (82M parameters) from 400 SFT samples, 296 KTO pairs, and 100 DPO pairs; the Lagrangian loop runs 30 steps at batch size 4 with KL coefficient 0.05, and the compliance threshold sweep covers $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$. SPECCOV counts substantive-token overlap with the source cluster, tokens of length ≥ 5 after stop-word filtering, with quoted-string and coverage bonuses.

Appendix B. Annotator Guidelines

The lead author (PhD in software engineering with prior mobile-app review-mining experience) served as primary annotator across all five annotation tasks. The 490-review classification gold was additionally annotated by two further independent human raters on the identical item set, aligned by review id; all three first completed a shared 20-review calibration round, disagreements were adjudicated before the main task opened, and the released gold label is the majority vote over the three raters. All human annotation in this work is carried out by those same three people. For Stage 4, all three raters scored all 200 rows of the reported re-run independently and blind to condition; the discarded 400-row round had the

lead author as primary rater with one additional rater on a 30-pair subset, and is described here only because the paper reports its withdrawal. Separately, two Qwen2.5-3B-Instruct prompts (concise role-based and chain-of-thought) served as the second and third raters in the 99-review cluster-purity Krippendorff α calibration, which is an LLM-assisted panel and not one of the human rounds. For the 5,230-record verified anchor, which covers 5,229 distinct corpus rows, annotators classified each review into one of seven Maalej-Nabil classes (**bug_report**, **feature_request**, **performance**, **usability**, **compatibility**, **praise**, **other**) with multi-label assignment permitted when a review carried multiple concerns; reviews below the “fairly confident” threshold were skipped and excluded from the anchor. For the 490-review classification gold standard, a stratified sample of 70 reviews per class was drawn from the working corpus, and the lead author re-labeled each review independently and blinded to the V2 LLM prediction; the 490 gold labels serve as the held-out evaluation set for Cohen’s κ progression (Table 12) and per-class F1 (Table 13). For the 5-dimension IssueSpec rubric (all 100 specs; an earlier stratified $n = 28$ subsample, 6 per issue type plus 2 extras), each spec was scored on a 1–5 Likert scale across five dimensions: *completeness* (are all required template fields filled?), *accuracy* (do field values correctly reflect the source cluster?), *actionability* (would a developer know what to do?), *specificity* (does the spec name concrete entities rather than generic terms?), and *clarity* (is the spec easy to read and unambiguous?), with each dimension scored independently. For the reported Stage 4 evaluation (100 reviews $\times$ 2 conditions, the same model with and without the IssueSpec, 200 rows), all three raters rated *quality* (1–5), *specificity* (1–5), and *helpful* (Y/N) while blinded to the generation method; the superseded 400-row round used four template-composed conditions and is retained in the release for comparison. In both rounds condition labels were replaced with random A/B/C/D codes and unsealed only at aggregation, with the assignment persisted as spreadsheets with JSON master keys for audit. A second human rater independently scored a 30-pair (**no_spec**, **full**) subset for inter-rater calibration. For the Y/P/N cluster-purity rubric (initial 50-cluster audit; 99-cluster three-rater subsample), annotators read 5 representative reviews per cluster and assigned *Y* (all 5 share the same sub-theme within the cluster’s aspect), *P* (3 or 4 of 5 share the sub-theme), or *N* (fewer than 3 share); weighted purity is $\text{purity}_w = (|Y| + 0.5|P|)/(|Y| + |P| + |N|)$, with *Y* contributing 1.0, *P* contributing 0.5, and *N* contributing 0. All comparative evaluations were blinded throughout the rating window and un-

1875 sealed only at aggregation; reviews or specs below the annotator confidence threshold were marked as “skip” and excluded from downstream analysis.

Table 5: IssueSpec standardized issue schema. Each issue type follows the dominant industry or HCI standard. Each template is drawn from a published standard or reference and reproduces its named fields; we ran no conformance check against the standards documents themselves, so “drawn from” should not be read as “certified against”.

Issue type	Template (citation)	Justification
bug_report	title, severity, steps_to_reproduce, expected, actual, component	The tracker-standard field set in GitHub Issues and JIRA. Chaparro et al. [4] support three of the six (observed behaviour, expected behaviour, steps to reproduce). We previously attributed the whole set to Zimmermann et al. [5]; that was wrong twice over. They propose no template, and in their developer survey severity ranks last at 0% importance and component at 3%, against 83% for steps to reproduce. We keep both fields because trackers require them, not because developers report finding them useful
feature_request	John user-story [40] (As-a / I-want / So-that) + behavior-driven development (BDD) acceptance criteria [41]	Dominant industry format (Agile, Scrum, SAFe); BDD standardizes testable outcomes
performance	Fields are latency, startup_time, memory_consumption, battery_drain, network_usage, mapped onto ISO/IEC 25010 [42] performance-efficiency sub-characteristics: the first two to time behaviour, the last three to resource utilization; capacity is not covered	International standard for software-quality NFR taxonomy. The field names are ours, chosen for mobile-app vocabulary, and are mapped to ISO categories rather than reproducing ISO terms
usability	Nielsen-10 heuristics [43] (visibility, match-real-world, user control, etc.)	Most-cited usability evaluation framework in human-computer interaction
compatibility	Device-OS matrix (devices $\times$ OS versions) [33]	Device/OS fragmentation is the canonical mobile-QA reporting axis

Table 6: Annotator-vs-LLM and inter-annotator deviation. The $\approx 25\%$ deviation is measured on the human-verified anchor; the κ -progression reaches moderate agreement [51]; the V5 endorsement row is a consistency check, not independent validation, since V5 trained on those corrections; LLM-judge calibration Δ bounds judge bias.

Annotator pair	sample	agreement / deviation
Lead-author vs V2 LLM	5,230	75% / 25% mis-labeled
Lead vs V2 / cleanlab / V5	489 gold	κ : 0.163 $\rightarrow$ 0.333 $\rightarrow$ 0.592
3 human raters, Y/N verdict	490 gold	Fleiss $\kappa =$ 0.968 ; 479/490 unanimous on the verdict, 476/489 on the label
3 human raters, 7-class label	489 gold	Fleiss $\kappa =$ 0.979 ; $\alpha = 0.979$
V2 LLM vs V5; cleanlab vs V5	215,583	71.96% / 86.77% agree
cleanlab vs V5 (consistency, not independent)	40,291	88.66% support
V5 vs gold, held out (replay-dependent, see §5.1)	307	$\kappa =$ 0.616
3 human raters (Stage 4 quality)	400 rated rows	weighted κ 0.970–0.986; within-1 100%
3 human raters (Stage 4 helpful)	400 rated rows	κ 0.678–0.836; $\alpha = 0.781$
3-rater <i>LLM-assisted</i> classification check (lead + 2 Qwen prompts)	99 re-views	Krippendorff $\alpha = 0.285$
Lead vs Qwen-judge (purity, rubric)	50 / 28	$\Delta = -0.035$ / +0.23

Table 7: What the experiments establish, in one line each. Negative findings are marked, and they are findings rather than caveats.

	finding	evidence
F1	Type routing, not the language model, is what produces tracker-ready fields: removing it collapses strict fill from 0.96 to 0.69 and the per-type structural metrics to zero.	§5.1, A3
F2	<i>Corrected.</i> We reported that LLM annotation noise concentrates on minority classes. Per stratum it does the opposite: 1,302 of 1,307 disagreements are in the largest class. What holds is that the labeller omits two classes entirely, which is the failure a verified anchor repairs.	§5.1
F3	<i>Negative, and it overturns our own earlier claim.</i> With the generator held fixed, adding the typed IR changes response quality by +0.04 ($p = 0.54$); the +2.35 we previously reported was an artifact of comparing two different text composers.	§5.2
F4	Generator capability shows up mostly as whether fields get filled, not as the quality of the text once filled: template-fill spans 0.96–0.48 across four models while rubric quality spans 4.09–3.77.	§5.1
F5	<i>Negative.</i> The knowledge graph loses to count-matched flat clustering on all three intrinsic metrics; its defensible role is centrality-based prioritization.	§5.4, A1b
F6	<i>Negative.</i> Extractive coverage does not measure faithfulness: SPECCOV shows no rank correlation with human judgement and orders generators backwards.	§5.1
F7	<i>Negative, then partial.</i> A CMDP constraint cannot be tested on a policy incapable of violating it; with one that can, the constraint binds and λ grows, but no trade-off is demonstrated.	§5.3
F8	Run-to-run generation noise is about 0.03 on template-fill, which is the yardstick separating the differences we claim from the ones we do not.	§6.4

Table 8: Cross-family five-dimension rubric, same Qwen2.5-3B judge as Table 10, on matched clusters. Panel A is the four-way comparison on clusters covered by every generator (bug reports only). Panel B widens the Claude-versus-Llama comparison to all five issue types. The Panel-A ordering matches the template-fill ordering in Table 9, but none of its six pairwise differences is significant at $n = 14$ (Wilcoxon p from 0.18 to 0.75), so read it as descriptive. Panel B widens the comparison to all five issue types and gives a nominal +0.38 for Claude over Llama at Wilcoxon $p = 0.015$; we withdraw that as a claim, because it fails Bonferroni or Holm correction over the seven tests in this table ($p = 0.104$) and an exact sign test ($p = 0.078$). Read both panels as descriptive.

generator	n	mean	compl.	accur.	action.	spec.	clarity
<i>Panel A, four-way, 14 matched bug-report clusters</i>							
Claude (Anthropic)	14	4.09	4.00	4.50	3.71	4.36	3.86
Llama-3.3-70B (Meta)	14	3.96	3.86	4.50	4.07	3.86	3.50
Qwen2.5-3B (Alibaba)	14	3.83	3.07	4.50	4.00	3.86	3.71
Qwen2.5-1.5B (Alibaba)	13	3.77	3.00	4.15	4.46	3.62	3.62
<i>Panel B, Claude vs Llama, 28 clusters, all five issue types</i>							
Claude (Anthropic)	28	4.01	3.86	4.54	3.75	4.29	3.64
Llama-3.3-70B (Meta)	28	3.64	3.39	4.46	3.14	3.82	3.36

Table 9: Multi-LLM evidence summary across three model families (Anthropic, Meta, Alibaba). Cross-family rank preservation in Stage 3 is the structural-bias control; LLM-as-judge calibration Δ vs the lead author bounds judge bias. *Two rows carry caveats a reader needs.* The Stage-4 row previously reported 0.61 versus 0.54 for “Claude” against Qwen-3B. We withdraw it: the file labelled Claude is `responses_rrgen_baseline.json`, which our own release notes record as the output of a deterministic template composer rather than of any language model, so the row compared a template against a generation. This is the same confound we withdrew in Experiment 2 and we had not noticed it here. The Stage-3 row is a real cross-family comparison but is not matched: Claude is scored on 100 clusters and Llama on 99, both spanning five issue types, whereas both Qwen conditions are 15 clusters of `bug_report` only. The 0.96-versus-0.48 span therefore compares a five-type mean against a single-type mean and should be read as an ordering, not as a magnitude.

Use		Model (family)	Result
Stage 3 generate		Claude (Anthropic) / Llama-3.3-70B (Meta) / Qwen-3B (Alibaba) / Qwen-1.5B (Alibaba)	0.96 / 0.80 / 0.74 / 0.48
Stage 4 generate		<i>withdrawn</i> (see <i>withdrawn</i> note)	
5-dim judge	rubric	Qwen-3B, all specs	100 3.94/5 (28-spec subsample: 3.89)
Cross-family rubric		Claude / Llama / Qwen-3B / Qwen-1.5B	4.09 / 3.96 / 3.83 / 3.77 (Table 8)
Y/P/N judge	purity	Qwen-3B vs lead-author	0.625 vs 0.66; $\Delta = -0.035$

Table 10: Experiment 1 five-dimension rubric (Qwen-judge, 1–5 scale) on the *complete* spec set: all 100 taxonomy specs and all 20 lead-author reference specs, scored in one run so the two columns are directly comparable. The middle column repeats the earlier stratified 28-spec subsample under the same judge and rubric; the gap between it and the full set is +0.05, which is the subsample error. The LLM-as-judge protocol follows the calibration-against-reference convention of [44, 45, 46].

dimension	taxonomy, $n = 100$	taxonomy, $n = 28$	lead-author ref, $n = 20$
completeness	3.72	3.79	3.20
accuracy	4.43	4.39	4.10
actionability	3.86	3.71	2.85
specificity	4.15	4.07	4.05
clarity	3.52	3.50	3.05
overall mean	3.94	3.89	3.45
sd across specs	0.59	—	0.41

Table 11: Experiment 1 substantive template-fill (strict criteria). The 0.96 vs 0.53 LLM-vs-GitHub gap is coverage, not content quality; the strict-vs-very-strict sweep preserves ordering. The exact-zero cells in **bugs:repro** and **features:story** columns for (b), (c), (d), and (e) are structural: free-form LLM, raw concatenation, lead-author, and human-GitHub specs do not emit the template-specific reproduction-step list or As-a/I-want/So-that triple at all when not prompted for them, so the strict-fill rate is zero by construction.

condition	strict	bugs:repro	features:story
(a) LLM + taxonomy	0.96	0.73	0.97
(b) LLM free-form	0.69	0.00	0.00
(c) raw concat	0.34	0.00	0.00
(d) lead-author ref ($n = 20$)	0.69	0.00	0.00
(e) human GitHub (3 repos)	0.53	0.13	0.00

Table 12: Cohen’s κ progression vs the 490-review expert gold. Three Landis-Koch [51] bands crossed; cleanlab correction follows the confident-learning procedure of [23]. The final column repeats the measurement against the majority-vote gold of the three-rater human panel, showing the trajectory is not an artifact of a single annotator.

classifier	n	Cohen’s κ	Landis-Koch band	κ vs 3-rater gold
V2 LLM original	489	0.163	slight	0.165
cleanlab-corrected V2	489	0.333	fair	0.334
V5 classifier	489	0.592	moderate	0.590

Table 13: Per-class F1 against the 490-review expert gold. V2 LLM is blind to the two minority classes (compatibility, performance). The verified-anchor procedure with targeted augmentation recovers them.

class	V2 LLM	cleanlab-V2	V5
bug_report	0.42	0.45	0.58
feature_request	0.32	0.37	0.57
performance	0.00	0.47	0.77
usability	0.11	0.27	0.55
compatibility	0.00	0.00	0.83
praise	0.38	0.62	0.68
other	0.30	0.47	0.60
macro F1	0.22	0.38	0.65

Table 14: Stage-4 re-run, same model with and without the IssueSpec, 100 paired reviews, three raters, all 200 rows scored by each. All figures are unadjusted, so the gain column equals the difference of the two condition columns; slot-stratified values are in the released JSON and differ by at most 0.03. The pooled effect is null, and the disagreement between the lead author and the two independent raters is larger than the effect being measured.

rater	no_spec	full	gain	95% CI	Wilcoxon p
lead author	4.04	4.60	+0.56	[+0.44, +0.68]	2×10^{-11}
independent rater 1	3.87	3.58	-0.29	[-0.56, -0.01]	0.06
independent rater 2	3.57	3.42	-0.15	[-0.33, +0.03]	0.09
pooled	3.83	3.87	+0.04	[-0.10, +0.18]	0.54

Table 15: Stage 2 designs head-to-head, comparing Pure HDBSCAN (without UMAP), original flat UMAP+HDBSCAN, and KG-hierarchical. Intrinsic metrics (Davies-Bouldin [58], Calinski-Harabasz [59], silhouette [60]) follow standard IR/clustering convention [61, 62, 63]. The Y/P/N weighted-purity rubric is a discrete-coding instantiation of the extrinsic cluster-purity metric [64, 61]. The count-controlled flat baseline (agglomerative-605) is reported separately in the A1b ablation (§5.5).

metric	Pure HDB	Flat U+H	KG hier.
n_clusters	3 (deg.)	194	605
mean cluster size	~16,933	471.0	15.9
Davies-Bouldin ($\downarrow$)	n/a	<i>withdrawn</i>	2.24
Calinski-Harabasz ($\uparrow$)	n/a	<i>withdrawn</i>	1.85
Y/P/N purity (lead-author)	n/a	0.66	not run
Y/P/N purity (Qwen-judge)	n/a	0.625	0.390

Table 16: Ablation summary. “Removing this hurts” is the only outcome that justifies a component. Three of the seven do not clear that bar, and we report them as findings.

id	what is removed	effect of removing it	verdict
A1	the KG, feeding flat clusters to Stage 3	Stage 3 quality unchanged	no effect
A1b	the KG, against flat clustering matched to the same 605 clusters	flat wins on all three intrinsic metrics	against the KG
A2	hierarchical sub-clustering	purity 0.625 flat vs 0.390 KG	against the KG
A3	taxonomy grounding, so the LLM writes free-form	strict fill 0.96 $\rightarrow$ 0.69; per-type structure to zero	justifies routing
A4	HITL-2 review of specs	no change to benchmark scores	no effect at benchmark time
A5	retrieval, keeping the IssueSpec	all automatic metrics move $< 1\%$	against RAG
A6	the IssueSpec, keeping retrieval	quality 3.87 $\rightarrow$ 3.83 ($-0.04, p = 0.54$)	no measurable effect (§5.2)
A7	single-stream feedback	RLHF not run	deferred